Year 5 Mathematics Solutions

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



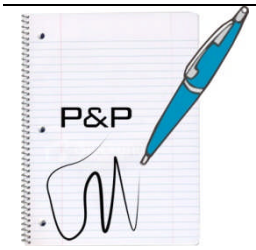
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



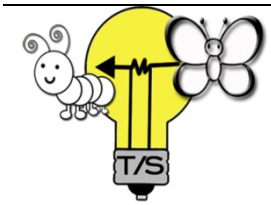
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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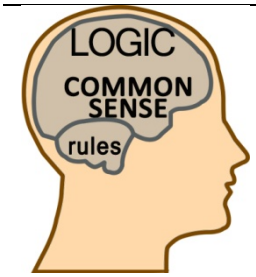
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

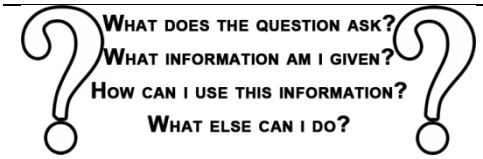
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 5 Mathematics

Number

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Exercise 1

Roman Numerals

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1) Convert the following Roman numerals to Arabic

a) V = 5

b) X = 10

c) C = 100

d) D =500

e) L = 50

2) Convert the following to Roman numerals

a) 10 = X

b) 200 = 100 + 100 = CC

c) 6= 5 + 1 = VI

d) 11 = 10 + 1 = XI

e) 105 = 100 + 5 = CV

3) Convert the following to Arabic numerals

a) LV = 50 + 5 = 55

b) CXI = 100 + 10 + 1 = 111

c) CLVII = 100 + 50 + 5 + 1 + 1 =157

d) XX = 10 + 10 = 20

e) LXXIII = 50 + 10 + 10 + 1 + 1 + 1 = 73

4) Convert the following to Roman numerals

a) 33 = 10 + 10 + 10 + 1 + 1 + 1 = XXXIII

b) 56 = 50 + 5 + 1 = LVI

c) 105 = 100 + 5 = CV

d) 12 = 10 + 1 + 1 = XII

e) 171 = 100 + 50 + 10 + 10 + 1 = CLXX1

5) Convert the following to Arabic numbers

a) XXIV = 10 + 10 + (5 – 1) = 24

b) LIX = 50 + (10 – 1) = 59

c) XCIX = (100 – 10) + (10 – 1)=99

d) CCIX = 100 + 100 + (10 – 1) = 209

e) XIX = 10 + (10 – 1) = 19

6) Convert the following to Roman numerals

a) 179 = 100 + 50 + 10 + 10 + (10 – 1) = CLXXIX

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b) 14 = 10 + (5 – 1) = XIV

c) 77 = 50 + 20 + 5 + 1 + 1 = LXXVII

d) 86 = 50 + 10 + 10 + 10 + 5 + 1 = LXXXVI

e) 111 = 100 + 10 + 1 = CXI

7) Which number between 1 and 100 would be the longest Roman numeral?

Since numbers in the forties and nineties are shown in the form XL..., or XC..., the required number must be in the thirties or eighties.

Numbers in the thirties are shown in the form XXX...

Numbers in the eighties are shown in the form LXXX...

Therefore the number must be in the eighties

Again, 9 is shown as IX, therefore the required unit place value must be 8

Therefore the number is 88; which is shown as LXXXVIII

8) Which number would be the first that requires four different characters in Roman numerals?

The first four different characters in Roman numerals are: I, V, X, and L

Since the X must be next to the L, the number must be of the form XL..., or LX...

Similarly, I and V must be shown as IV or VI

Since XL < LX, and IV < VI, the number is XLIV = 44

9) Write a Roman numeral that contains more than one different character and is a palindrome

The number requires at least two characters. The first two characters are I and V. The number could be IVI or VIV, neither of which are valid Roman numerals.

The next possible pair is I and X. The number could be IXI or XIX.

Of the two, XIX (= 19) is a valid Roman numeral, and therefore is the correct answer

10) Which of the following Roman numerals is incorrect? Give the correct Roman numeral.

a) 40 = XXXX

Incorrect: 40 = 50 – 10 = XL

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b) 99 = IC

Incorrect: 99 = (100 – 10) + (10 – 1) = XCIX

c) 95 = VC

Incorrect: 95 = (100 – 10) + 5 = XCV

d) 19 = IXX

Incorrect: 19 = 10 + (10 – 1) = XIX

e) 49 = XLIX

Correct

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Exercise 2

Place Value

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1) Write the following in numerals

a) Three hundred and twenty seven

= 3 x one hundred, and 2 x ten, and 7 = 327

b) Four thousand two hundred and twelve

= 4 x one thousand, and 2 x one hundred and 1 x ten and 2 = 4212

c) Seven hundred and seven

= Seven hundred and zero tens and seven = 797

d) Six thousand and fifteen

=Six thousand and zero hundreds and one ten and five = 6015

e) Twelve thousand four hundred and twenty

= 1 x Ten thousand and 2 x one thousand and 4 x 100, 2 x ten and zero = 12420

f) Thirty two thousand and eleven

= 3 x ten thousand, and 2 x one thousand, and no thousands, and no hundreds and 1 x 10 and 1 = 32011

2) Write the following in words

a) 3233

3 x one thousand, and 2 x one hundred, and 3 x ten and 3

= Three thousand 2 hundred and thirty three

b) 41002

= 4 x ten thousand, and 1 x one thousand, and no hundred, and no tens, and 3

= Forty one thousand and two

c) 706

= 7 x one hundred, and no tens and 6

= seven hundred and six

d) 5007

= 5 x one thousand, and no hundreds and no tens and 7

= Five thousand and seven

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e) 30207

= 3 x ten thousand, and no thousands, and 2 x one hundred, and no tens and 7

= Thirty thousand two hundred and seven

f) 100001

= 1 x one hundred thousand, and no ten thousands, and no thousands, and no hundreds, and no tens and 1

= One hundred thousand and one

3) What is the place value of the 5 in each of the following?

a) 1005

5 x one = 5

b) 51443

5 x ten thousand = 50,000

c) 75111

5 x one thousand = 5000

d) 523123

5 x one hundred thousand = 500,000

e) 54

5 x ten = 50

f) 65121

5 x one thousand = 5000

4) Write the following numbers in order, from largest to smallest

121234, 11246, 13652, 834, 999, 1011, 1101,

Look for largest numbers by comparing the same place values from left to right

Three digit numbers must be smallest

834<999

Then 4 digit numbers

1011<1101

Then 5 digit numbers

11246<13652

Order is 121234, 13652, 11246, 1101, 1011, 999, 834

5) Write the following numbers in order, from smallest to largest

4224, 425, 501, 5001, 516, 111, 1111, 11002, 1009

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As for question 4:

Three digit numbers are smallest

111<425<501<516

Then four digit numbers

1009<1111<4224<5001

Order is 111, 425, 501, 516, 1009, 1111, 4224, 5001, 11002

6) There were 26244 people at a soccer match. Write this number to the nearest

a) Hundred

244 to the nearest hundred is 200

Rounded number is 26200

b) Thousand

6244 to the nearest thousand is 6000

Number is 26000

c) Ten thousand

26244 to the nearest ten thousand is 30000

7) Round the number 67532556 to the nearest:

a) Ten

56 to the nearest ten is 60

Number is 67,532,560

b) Hundred

556 to the nearest hundred is 600

Number is 67,532,600

c) Thousand

2556 to the nearest thousand is 3000

Number is 67,533,000

d) Ten thousand

32556 to the nearest ten thousand is 30000

Number is 67,530,000

e) Hundred thousand

532556 to the nearest hundred thousand is 500000

Number is 67,500,000

f) Million

7532556 to the nearest million is 8,000,000

Number is 68,000,000

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8) Add the following

a) 327 + five hundred and seventy five

$$\begin{array}{r} 327 \\ +\ 575 \\ \hline 902 \end{array}$$

b) Two thousand and nine + 747

$$\begin{array}{r} 2009 \\ +\ 747 \\ \hline 2756 \end{array}$$

c) Twenty thousand one hundred + eighteen thousand two hundred and twelve

$$\begin{array}{r} 20{,}118 \\ +18{,}212 \\ \hline 38{,}340 \end{array}$$

d) 1143 + three thousand one hundred and two

$$\begin{array}{r} 1143 \\ +3102 \\ \hline 4245 \end{array}$$

e) 17111 + three hundred and ninety nine

$$\begin{array}{r} 17111 \\ +\ 399 \\ \hline 17510 \end{array}$$

9) Which numeral represents hundreds in the number 323468

8 represents ones

6 represents tens

4 represents hundreds

10) If 50,000 is added to the number 486,400, which numerals change place value?

$$\begin{array}{r} 486400 \\ +50000 \\ \hline 536400 \end{array}$$

The 4 representing hundreds of thousands changes to a 5

The 8 representing tens of thousands changes to a 3

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Exercise 3

Factors & Multiples

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1) List the factors of the following numbers

a) 7

Only 1 and 7 divide evenly into 7. (Numbers with only 2 factors are prime)

b) 9

1, 3 and 9 (numbers with an odd number of factors are square numbers)

c) 10

1, 2, 5, 10

d) 12

1, 2, 3, 4, 6, 12

e) 25

1, 5, 25 (square number)

f) 30

1, 2, 3, 5, 6, 10, 15, 30

2) By using a factor tree find the prime factors of the following

NOTE: 1 is NOT a prime number

a) 16

16 = 2 x 8

8 = 2 x 4
4 =

Therefore 8 = 2 x 2 x 2

2 is the only prime factor of 8

b) 20

20 = 2 x 10

10 = 2 x 5

Therefore 20 = 2 x 2 x5

2 and 5 are the prime factors of 20

c) 64

64 = 2 x32

32 = 2 x 16

16 = 2 x 8

8 = 2 x 4

4 = 2 x 2

Therefore 64 = 2 x 2 x 2 x 2 x 2 x 2 x 2 x 2

2 is the only prime factor of 64

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d) 100

100 = 2 x50

50 = 2 x 25

25 = 5 x 5

Therefore 50 = 2 x 5 x 5

2 and 5 are the prime factors of 100

e) 144

144 = 2 x 72

72 = 2 x 36

36 = 2 x 18

18 = 2 x 9

9 = 3 x 3

Therefore 144 = 2 x 2 x 2 x 2 x 3 x 3

2 and 3 are the prime factors of 144

f) 261

261 = 3 x 87

87 = 3 x 29

29 is a prime number

Therefore 261 = 3 x 3 x 29

3 and 29 are the prime factors of 261

3) Find the greatest common factor of the following pairs of numbers

a) 2 and 6

The factors of 2 are 1 and 2

The factors of 6 are 1, 2, 3, and 6

The GCF of 2 and 6 is 2

b) 6 and 15

The factors of 6 are 1, 2, 3, and 6

The factors of 15 are 1, 3, 5, and 15

The GCF is 3

c) 10 and 25

The factors of 10 are 1, 2, 5, and 10

The factors of 25 are 1, 5, and 25

The GCF is 5

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d) 14 and 49

The factors of 14 are 1, 2, 7, and 14

The factors of 49 are 1, 7, and 49

The GCF is 7

e) 12 and 64

The factors of 12 are 1, 2, 3, 4, 6, and 12

The factors of 64 are 1, 2, 4, 8, 16, 32, and 64

The GCF is 4

f) 36 and 99

The factors of 36 are 1, 2, 3, 4, 6, 9, 12, 18, and 36

The factors of 99 are 1, 3, 9, 11, 33, and 99

The GCF is 9

4) List all the multiples of the following that are less than 50

a) 3

3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48

b) 4

4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48

c) 5

5, 10, 15, 20, 25, 30, 35, 40, 45

d) 7

7, 14, 21, 28, 35, 42, 49

e) 10

10, 20, 30, 40

f) 15

15, 30, 45

5) List the multiples of the following that are greater than 50 and less than 75

a) 2

52, 54, 56, 58, 60, 62, 64, 66, 68

b) 5

55, 60, 65

c) 6

54, 60, 66

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d) 8

56, 64

e) 11

55, 66

f) 40

There are no multiples of 40 between 50 and 70

6) Find the least common multiple of the following pairs of numbers

a) 2 and 3

Multiples of 2 are 2, 4, 6, 8, ...

Multiples of 3 are 3, 6, 9, 12, ...

LCM is 6

b) 3 and 5

Multiples of 3 are 3, 6, 9, 12, 15,.....

Multiples of 5 are 5, 10, 15, 20, ...

LCM is 15

c) 4 and 6

Multiples of 4 are 4, 8, 12, 16, ...

Multiples of 6 are 6, 12, 18, 24, ...

LCM is 12

d) 5 and 20

Multiples of 5 are 5, 10, 15, 20, 25, ...

Multiples of 20 are 20, 40, 60, ...

LCM is 20

e) 6 and 32

Multiples of 6 are 6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90, 96, 102, ...

Multiples of 32 are 32, 64, and 96

LCM is 96

f) 10 and 12

Multiples of 10 are 10, 20, 30, 40, 50, 60, ...

Multiples of 12 are 12, 24, 36, 48, 60, and 72

LCM is 60

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7) Jim writes the letter X on every 8th page of a book, while Tony writes the letter A on every 10th page.

a) What is the first page that has an X and an A?

The letter X will appear on pages 8, 16, 24, 32, 40, 48, ...

The letter A will appear on pages 10, 20, 30, 40, 50, ...

The first page with both letters will be page 40

b) What are the first 3 pages that have an X and an A on them?

Continuing the pattern, the next page will be 80

The letters appear together every 40 pages, therefore the next 3 page numbers are 40, 80, and 120

c) If the book has 300 pages what is the last page in the book that has an X and an A?

Continuing the pattern, the letters will appear on pages 160, 200, 240, 280

8) A stamp collector has 24 Australian stamps, 40 English stamps, and 64 American stamps. If each page of his album has the same number of stamps, how many stamps are on each page, and how many pages are in the album? Note the stamps of different countries cannot be on the same page.

If each page has the same number of stamps, this number must divide evenly into all three numbers in the question. In other words, we are looking for the factors of the three numbers.

The factors of 24 are 1, 2, 3, 4, 6, 8, 12, and 24

The factors of 40 are 1, 2, 5, 8, 20, and 40

The factors of 64 are 1, 2, 4, 8, 16, 32, and 64

The common factors of the three numbers are 1, 2, and 8

The album could have 1 stamp on each page and have 128 pages

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The album could have 2 stamps on each page and have 64 pages

The album could have 8 stamps on each page and have 13 pages

9) A loaf of bread contains 24 slices and a packet of ham has 5 slices. What is the smallest number of loaves of bread and packets of ham that must be bought to make **sandwiches** so there is no bread or ham left over? How many sandwiches will be made?

There are 2 slices of bread in a sandwich, so there are 12 sandwich pairs in a loaf

If we buy 1 loaf we have 12 pairs, 2 loaves give us 24 pairs, 3 loaves gives us 36 pairs, 4 loaves give us 48 pairs, and 5 loaves gives us 60 pairs.

Every pair of breads must have 1 slice of ham, therefore the number of pairs of bread must be a multiple of 5 in order to use up all the ham

The LCM of 12 and 5 is 60

Therefore we must buy 12 packets of ham, 5 loaves of bread to make 60 sandwiches

10) A light flashes every 6 seconds, and a horn sounds every 9 seconds. In two minutes how many times will the light flash and the horn sound at the same time?

The light will flash after the following number of seconds: 6, 12, 18, 24, 30, 36, 42, ...

The horn will sound after 9, 18, 27, 36, ... seconds

They will occur at the same time after 18 and 36 seconds. Continuing the pattern up to 120 seconds (2 minutes) gives 18, 36, 54, 72, 90, 108 seconds

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Exercise 4

Operations on Whole Numbers

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1) Add the following

a) 54 + 26

$$\begin{array}{r} 54 \\ +\ 26 \\ \hline 80 \end{array}$$

b) 17 + 47

$$\begin{array}{r} 17 \\ +\ 47 \\ \hline 64 \end{array}$$

c) 21 + 45

$$\begin{array}{r} 21 \\ +\ 45 \\ \hline 66 \end{array}$$

d) 19 + 55

$$\begin{array}{r} 19 \\ +\ 55 \\ \hline 74 \end{array}$$

e) 33 + 62

$$\begin{array}{r} 33 \\ +\ 62 \\ \hline 95 \end{array}$$

f) 72 + 22

$$\begin{array}{r} 72 \\ +\ 22 \\ \hline 94 \end{array}$$

2) Subtract the following

a) $99 - 54$

$$\begin{array}{r} 99 \\ -\ 54 \\ \hline 45 \end{array}$$

b) $83 - 32$

$$\begin{array}{r} 83 \\ -\ 32 \\ \hline 51 \end{array}$$

c) $67 - 46$

$$\begin{array}{r} 67 \\ -\ 46 \\ \hline 21 \end{array}$$

d) $71 - 51$

$$\begin{array}{r} 71 \\ -\ 51 \\ \hline 20 \end{array}$$

e) $84 - 13$

$$\begin{array}{r} 83 \\ -\ 13 \\ \hline 70 \end{array}$$

f) $57 - 45$

$$\begin{array}{r} 57 \\ -\ 45 \\ \hline 12 \end{array}$$

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3) Add the following

a) 93 + 68

$$\begin{array}{r} 93 \\ +\ 68 \\ \hline 161 \end{array}$$

b) 64 + 46

$$\begin{array}{r} 64 \\ +\ 46 \\ \hline 110 \end{array}$$

c) 73 + 51

$$\begin{array}{r} 73 \\ +\ 51 \\ \hline 124 \end{array}$$

d) 112 + 103

$$\begin{array}{r} 112 \\ +\ 103 \\ \hline 215 \end{array}$$

e) 146 + 119

$$\begin{array}{r} 146 \\ +\ 119 \\ \hline 265 \end{array}$$

f) 163 + 104

$$\begin{array}{r} 163 \\ +\ 104 \\ \hline 267 \end{array}$$

4) Subtract the following

a) $274 - 162$

$$\begin{array}{r} 274 \\ -\ 162 \\ \hline 112 \end{array}$$

b) $312 - 153$

$$\begin{array}{r} 312 \\ -\ 153 \\ \hline 159 \end{array}$$

c) $422 - 113$

$$\begin{array}{r} 422 \\ -\ 113 \\ \hline 209 \end{array}$$

d) $812 - 333$

$$\begin{array}{r} 812 \\ -\ 133 \\ \hline 679 \end{array}$$

e) $713 - 618$

$$\begin{array}{r} 713 \\ -\ 618 \\ \hline 95 \end{array}$$

f) $901 - 565$

$$\begin{array}{r} 901 \\ -\ 565 \\ \hline 336 \end{array}$$

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5) Multiply the following

a) 42×5

40 x 5 =200
2 x 5 = 10

42 x 5 = 210

b) 33×8

30 x 8 = 240
3 x 8 = 24

33 x 8 = 264

c) 7×52

7 x 50 = 350
7 x 2 = 14

7 x 52 = 364

d) 11×13

10 x 13 = 130
1 x 13 = 13

11 x 13 = 143

e) 27×12

20 x 12 = 240
7 x 12 = 84

27 x 12 = 324

f) 31×15

30 x 15 = 450
1 x 15 = 15

31 x 15 = 465

6) Multiply the following

a) 34×27

30 x 20 = 600
30 x 7 = 210
4 x 20 = 80
4 x 7 = 28

34 x 27 = 918

b) 52×28

50 x 20 = 1000
50 x 8 = 400
2 x 20 = 40
2 x 8 = 16

52 x 28 = 1456

c) 61×22

60 x 20 = 1200
60 x 2 = 120
1 x 22 = 22

61 x 22 = 1342

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d) 53×41

50 x 40 = 2000
50 x 1 = 50
3 x 40 = 120
3 x 1 = 3

53 x 41 = 2173

e) 66×37

60 x 30 = 1800
60 x 7 = 420
6 x 30 = 180
6 x 7 = 42

66 x 37 = 2442

f) 71×19

70 x 10 = 700
70 x 9 = 630
1 x 19 = 19

71 x 19 = 1349

7) Divide the following

a) $99 \div 9$

11 x 9 = 99

99 ÷ 9 = 11

b) $84 \div 7$

12 x 7 = 84

84 ÷ 7 = 12

c) $54 \div 6$

6 x 9 = 54

54 ÷6 = 9

d) $78 \div 12$

78 = 72 - 6

72 ÷ 12 = 6

6 ÷ 12 = 0.5

78 ÷ 12 = 6.5

e) $95 \div 4$

95 = 92 + 3

92 ÷ 4 = 23

3 ÷ 4 = 0.75

95 ÷ 4 23.75

f) $86 \div 8$

80 ÷ 8 = 10

6 ÷ 8 = 0.75

86 ÷ 8 = 10.75

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8) Divide the following

a) $150 \div 15$

150 = 15 x 10

150 ÷ 15 = 10

b) $220 \div 10$

220 = 22 x 10

220 ÷ 10 = 22

c) $180 \div 20$

180 = 18 x 10 = 9 x 20

180 ÷ 9 = 20

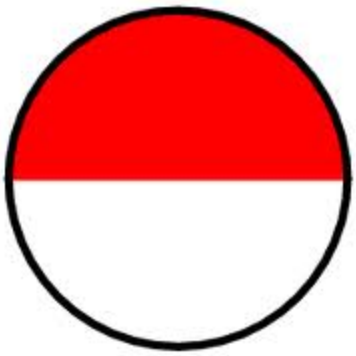
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Exercise 5

Unit Fractions: Comparison & Equivalence

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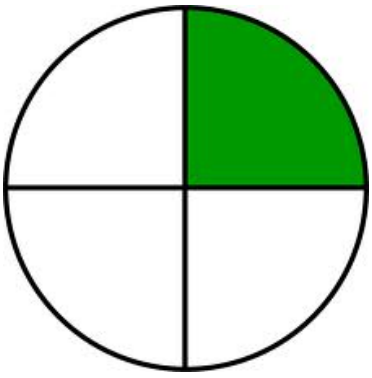
Use the following diagrams to help understand the solutions



$\frac{1}{2}$

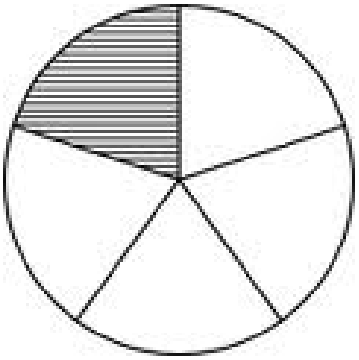


$\frac{1}{3}$

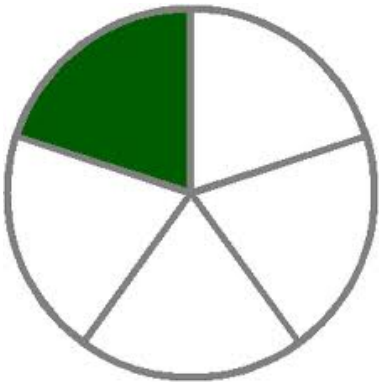


$\frac{1}{4}$

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$\frac{1}{5}$



$\frac{1}{6}$

Etcetera

1) Which is the bigger fraction?

a) $\frac{1}{2} > \frac{1}{5}$

b) $\frac{1}{7} < \frac{1}{4}$

c) $\frac{1}{5} > \frac{1}{6}$

d) $\frac{1}{3} > \frac{1}{8}$

e) $\frac{1}{20} < \frac{1}{10}$

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2) Put the following in order from largest to smallest

a) $\frac{1}{5}, \frac{1}{2}, \frac{1}{3}$

$\frac{1}{2} > \frac{1}{3} > \frac{1}{5}$

b) $\frac{1}{7}, \frac{1}{3}, \frac{1}{6}$

$\frac{1}{3} > \frac{1}{6} > \frac{1}{7}$

c) $\frac{1}{9}, \frac{1}{10}, \frac{1}{2}$

$\frac{1}{2} > \frac{1}{9} > \frac{1}{10}$

d) $\frac{1}{2}, \frac{1}{11}, \frac{1}{5}$

$\frac{1}{2} > \frac{1}{5} > \frac{1}{11}$

3) John eats one-third of a cake and Peter eats one-fifth. Who has more cake left?

$\frac{1}{5} < \frac{1}{3}$

Therefore Peter has eaten less cake and has more left

4) Debbie and Anne drive the same type of car and both go to the same petrol station at the same time to fill their petrol tanks. Debbie needs half a tank of petrol tank to be full, while Anne needs a quarter of a tank to fill up. Who will have to pay more for petrol

$\frac{1}{2} > \frac{1}{4}$

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Therefore Anne has to put more petrol in and will pay more

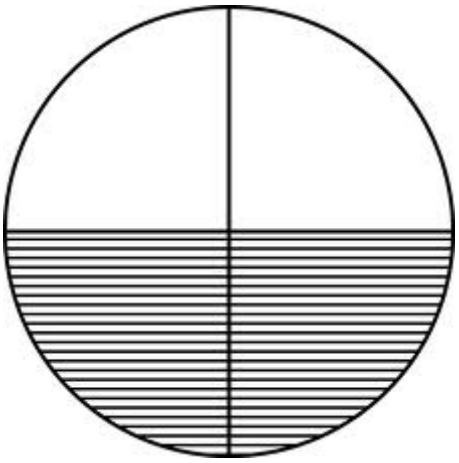
5) Bill and Ben start running at the same time. After one minute Bill has run one-quarter of a lap and Ben one-fifth of a lap. If they continue to run at the same speed, who will finish the lap first?

$\frac{1}{4} > \frac{1}{5}$

Therefore Bill has run further and will finish the lap first

6) Which of the following fractions is the fraction $\frac{1}{2}$ equal to?

$\frac{3}{5}, \frac{3}{6}, \frac{3}{7}, \frac{2}{4}, \frac{4}{10}$



The diagram shows a circle cut into quarters. Two of the quarters have been shaded in. It can be seen that this is the same as one half. Therefore $\frac{2}{4} = \frac{1}{2}$ Draw similar diagrams to show that :$\frac{3}{6} = \frac{1}{2}$, and that none of the other fractions are equal to one half

7) Four friends decide to share a pizza. If they each have an equal sized piece and eat all the pizza between them, what fraction of the pizza does each person get?

Four equal sized pieces take up $\frac{1}{4}$ of the pizza each. See diagram from Q6

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8) In a mathematics test Tom got $\frac{1}{4}$ of the questions wrong, and Alan got $\frac{1}{3}$ of the questions wrong. Who did better on the test?

$\frac{1}{3} > \frac{1}{4}$

Therefore Alan got more questions wrong and did worse on the test; Tom did better.

9) Josh and Tim are each reading a book. Josh's book has 10 chapters of which he has read 5, while Tim has read 4 out of 8 chapters. Who has read the greater fraction of their book?

$\frac{5}{10} = \frac{1}{2}$

$\frac{4}{8} = \frac{1}{2}$

Therefore they have read the same fraction of their book

10) Put the following fractions in order from smallest to largest

$\frac{1}{3}, \frac{2}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{6}, \frac{1}{9}$

$\frac{1}{9} < \frac{1}{4} < \frac{1}{3} < \frac{1}{2} = \frac{2}{4} = \frac{3}{6}$

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Exercise 6

Operations on Decimals: Money problems

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1) Order the following from smallest to largest

0.4, 0.25, 0.33, 0.11, 0.05, 0.9, 0.09, 0.5, 0.01, 0.1

Using place value:

0.01 < 0.05 < 0.09 < 0.1 <0.11 <0.25 < 0.33 <0.4 < 0.5 <0.9

2) Order the following from largest to smallest

0.91, 0.19, 1.34, 0.34, 0.09, 1.91, 0.03, 0.05, 0.55, 1.55, 0.195

Using place value:

1.91 > 1.55 > 1.34 > 0.91 >0.55 > 0.34 > 0.195 > 0.19 > 0.09 > 0.05 > 0.03

3) Add the following

a) 0.23 + 0.42

0.23
0.42
0.65

b) 0.15 + 0.62

0.15
0.62
0.77

c) 0.33 + 0.45

0.33
0.45
0.78

d) 0.71 + 0.28

0.71
0.28
0.99

e) 0.55 + 0.45

0.55
0.45
1.00

f) 0.8 + 0.3

0.8
0.3
1.1

4) Add the following

a) 0.58 + 0.36

0.58
0.36
0.94

b) 0.75 + 0.18

0.75
0.18
0.93

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c) 0.22 + 0.69

0.22
0.69
0.91

d) 0.54 + 0.87

0.54
0.87
1.41

e) 0.99 + 0.51

0.99
0.51
1.50

f) 0.86 + 0.48

0.86
0.48
1.34

5) Add the following

a) 1.42 + 2.11

1.42
2.11
3.53

b) 1.61 + 0.22

1.61
0.22
1.83

c) 2.35 + 1.21

2.35
1.21
3.56

d) 4.23 + 1.62

4.23
1.62
5.84

e) 5.11 + 3.11

5.11
3.11
8.22

f) 1.55 + 1.56

1.55
1.56
3.11

6) Add The following

a) 2.67 + 4.44

2.67
4.44
7.11

b) 3.68 + 3.54

3.68
3.54
7.22

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c) 2.59 + 4.62

2.59
4.62
7.21

d) 1.99 + 3.98

1.99
3.98
5.97

e) 6.77 + 3.25

6.77
3.25
10.02

f) 3.49 + 4.88

3.49
4.88
8.37

7) Subtract the following

a) 0.54 – 0.23

0.54
0.23
0.31

b) 0.86 – 0.13

0.86
0.13
0.73

c) 0.99 – 0.48

0.99
0.48
0.51

d) 0.77 – 0.66

0.77
0.66
0.11

e) 0.12 – 0.02

0.12
0.02
0.10

f) 0.25 – 0.24

0.25
0.24
0.01

8) Subtract the following

a) 1.41 – 0.61

1.41
0.61
0.80

b) 1.89 – 0.92

1.89
0.92
0.97

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c) 2.12 – 0.43

2.12
0.43
1.69

d) 3.24 – 2.56

3.24
2.56
0.68

e) 9.57 – 7.94

9.57
7.94
1.63

f) 2.15 – 0.99

2.15
0.99
1.16

9) Tom has $2.67 and lends Alan $1.41. How much money has Tom now got?

2.67
1.41
1.26

Tom has $1.26 left

10) Francis buys a pen for $1.12, a ruler for $0.46 and a book for $5.20. How much did he spend in total?

1.12
0.46
5.20
6.78

Francis spent $6.78

11) At a fast food place, burgers are $4.25, fries are $1.60, drinks are $1.85, and ice creams are $0.55 each. How much money is spent on each of the following?

a) A burger and fries

4.25
1.60
5.85

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b) A burger, drink and ice cream

4.25
1.85
0.55
6.65

c) Two burgers

4.25
4.25
8.50

d) Two fries and a drink

1.60
1.60
1.85
5.05

e) Two drinks and two ice creams

1.85
1.85
0.55
0.55
4.80

12) Martin gets $10 pocket money. He spends $1.65 on a magazine, $1.15 on a chocolate bar, $3.75 on food for his pet fish, and $1.99 on a hat. How much pocket money does he have left?

1.65
1.15
3.75
1.99
8.54

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10.00
 8.54
 1.46

Martin has $1.46 left

13) How much change from $20-should a man get who buys two pairs of socks at $2.50 each and a tie for $6.90?

 2.50
 2.50
 6.90
11.90

20.00
11.90
 8.10

The man will have $8.10 change

14) Peter needs $1.25 for bus fare home. If he has $5 and buys 3 bags of chips that cost $1.40 each, how much money does he have to borrow from his friend so he can ride the bus home?

1.40
1.40
1.40
4.20

5.00
4.20
0.80

1.25
0.80
0.45

Peter needs to borrow 45 cents to give him the $1.25 he needs for bus fare

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Year 5 Mathematics

Chance & Data

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Exercise 1

Simple & Everyday Events

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1) Put the following events in order from least likely to happen to most likely to happen

a) You will go outside of your house tomorrow

b) You will find a $100 note on the ground

c) The sun will rise tomorrow

d) You will pass a maths test you didn't study for

e) You will be elected President of the United States within the next year

f) You will toss a coin and it will land on heads

There is no chance that you can become President within a year even if you were eligible

There is only a small chance that you will find a $100 note

You will probably fail a maths test if you don't study for it, which is less than 50%

The chances of a coin landing on heads is 50% (1/2)

You will probably go outside at some stage tomorrow

The sun will definitely rise tomorrow

2) A boy's draw has 3 white, 5 black and 2 red t-shirts in it. If he reaches in without looking:

a) What colour t-shirt does he have the most chance of pulling out?

There are more black t-shirts; therefore he is most likely to pull a black one out

b) What colour t-shirt does he have least chance of pulling out?

There are less red t-shirts; therefore he is least likely to pull a red one out

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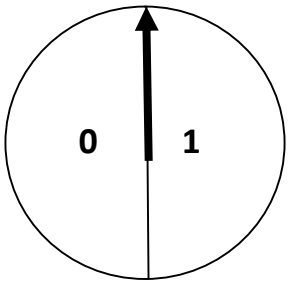
c) What chance does he have of pulling out a blue t-shirt?

Since there are no blue t-shirts in the draw, his chances of pulling one out are zero; it is impossible

3) A man throws a coin 99 times into the air and it lands on the ground on heads every time. Assuming the coin is fair, does he more chance of throwing a head or a tail on his next throw? Explain your answer

The chances of throwing a head (or a tail) are exactly the same on every throw. It is extremely unlikely that he has thrown a coin 99 times and got a head each time, but each individual throw has the same chance of coming up heads

4) A person spins the spinner shown in the diagram. If he does this twice and adds the two numbers spun together what total is he most likely to get?



On each spin he is equally likely to get a 0 or a 1. So on two throws he could get:

- A zero then another zero (total of 0)
- A zero then a 1 (total of 1)
- A 1 then a zero (total of 1)
- A 1 then another 1 (total of 2)

Therefore he has most chance of throwing a total of 1

5) A man has 2 blue socks and 2 white socks in a draw. If he pulls out a blue sock first, is he more likely or less likely to get a pair if he chooses another sock with his eyes closed?

After he pulls out the blue sock, he could pull the other blue, the first white sock, or the second white sock. He is more likely to pull out a white sock and therefore he is less likely to end up with a pair

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6) There are 10 blue, 10 green and 10 red smarties in a box. If a person takes one from the box without looking, which colour is he most likely to pull out? If he keeps pulling smarties out, how many smarties must he pull out in total to make sure he gets a green one

Since there is the same number of each colour, each colour has an equal chance of being pulled out at first.

It is possible that he pulls out all the red and all the blue before pulling a green one out. The only way he can be certain of getting a green one is if there are only green ones left. Therefore he must pull a total of 21 smarties to make sure he gets a green smartie

7) John thinks of a number between 1 and 10, while Alan thinks of a number between 1 and 20. Whose number do I have a better chance of guessing?

Since there are only 10 possible numbers to choose from in John's number, I have a better chance of guessing his correctly

8) A set of triplets is starting at your school tomorrow. You do not know how many of them are boys and how many are girls. List all the possible combinations they might be.

The possibilities are

BBB (three boys)
BBG (two boys and a girl)
BGG (two girls and a boy)
GGG (three girls)

NOTE: The possibilities do not have an equal chance of occurring. As an extension, which combination(s) is/are more likely?

9) Our school canteen has mini pizzas with three toppings on each one. The toppings are selected from:

- Ham (H)
- Pineapple (P)
- Anchovies (A)
- Olives (O)

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a) What are the possible combinations of pizza available?

The combinations are:

- HPA
- HPO
- HAO
- PAO

b) If I do not like anchovies, how many pizzas from part a will I like?

There is only one possible pizza that does not contain anchovies:

- HPO

c) If EVERY pizza MUST HAVE ham as one of the three toppings, how does this change the answers to questions a and b?

The only allowable combinations would be

- HPA
- HPO
- HAO

The option HPO without anchovies would still be available

10) On my lotto ticket I mark the numbers

1, 2, 3, 4, 5, 6

My friend's numbers are

12, 18, 19, 23, 27, 42

Which one of us is more likely to win Lotto? Explain your answer

Although it appears that my friend has a better chance of winning, my numbers have an equal chance of being chosen as his. Since the chances of me winning lotto with my numbers would be extremely small, the question shows how hard it is for anybody to win!

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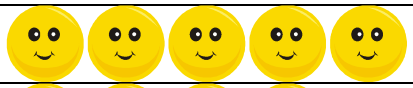
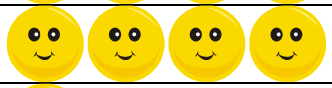
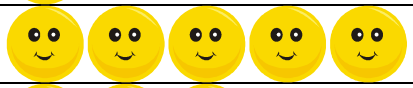
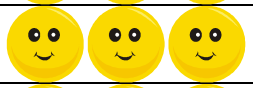
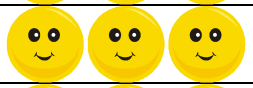
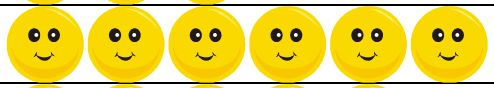
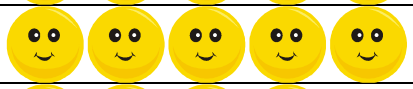
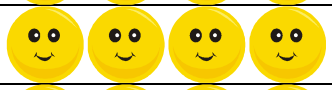
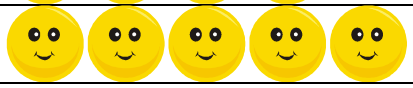
Exercise 2

Picture Graphs

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1) The picture graph below shows the approximate attendance at a soccer match for the past ten games

Each "face" represents 1000 people

Game Number	Attendance
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

a) For which game was there the largest crowd and what was the approximate attendance?

Game 7 had approximately 60,000 people

b) Which two consecutive games had approximately the same size crowd?

Games 5 and 6 had the same approximate crowd

c) What was the most common attendance figure?

Approximately 50,000 attended four times

d) For one game the weather was cold and windy and there was a transport strike. Which game number was this most likely to be? Approximately how many people attended this game?

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Game 3 only had approximately 1000 people attending, so it is most likely this was the game

2) The picture graph below shows the approximate number of fish caught at a beach over the past ten years. Each "fish" represents 500 fish

Year	Fish caught
2002	
2003	
2004	
2005	
2006	
2007	
2008	
2009	
2010	
2011	

a) Approximately how many fish were caught in 2003?

6 "fish" x 500 = 3000 fish

b) In which year were the most fish caught and how many was this?

In 2004 there are 8 "fish" x 500 = 4000 fish

c) In what year do you think the government put a restriction on the number of fish that could be caught?

The number of fish caught was a lot less in 2007

d) How many fish have been caught in total over the past ten years?

There are 49 "fish" x 500 = 24500 fish

3) The approximate average temperature for selected months for a city is shown in the picture graph below. Each  represents 10 degrees, each  represents 5 degrees

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Month	Average daytime temperature
February	
April	
June	
August	
October	
December	

a) Which are the hottest months of those shown?

February and December

b) Which are the coldest months of those shown?

June and August

c) What is the average temperature in October?

10 + 10 + 5 = 25 degrees

d) From this graph estimate the average temperature for this city in November

The average November temperature would be probably between 25 and 30 degrees.

e) From the graph, is this city in the northern or southern hemisphere? Explain your answer

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Southern hemisphere (summer in the December-February period, winter in the June-August period)

4) Jenny wanted to use a picture graph to show the number of people living in the 20 biggest cities in the world. Why would the following be a poor choice for a symbol?

 = 1 person

Propose a better choice

The graph would be far too large: a city having 20 million people would require 20 million symbols!

A better choice may be to have each symbol represent 2 million people

5) A class took a survey of each student's favourite fruit and drew the following graph from their results. One piece of fruit equals one vote



a) What is the most popular fruit in this class?

Bananas (5 votes)

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b) How many students' favourite fruit is watermelon?

Three

c) How many students are in the class?

There were 17 votes in the survey, so assuming no one was absent and everyone voted, there are 17 students in the class

d) The voting was from a list given to the students by their teacher. Nobody voted for a lemon as their favourite fruit. Discuss how this shows limitations of using picture graphs

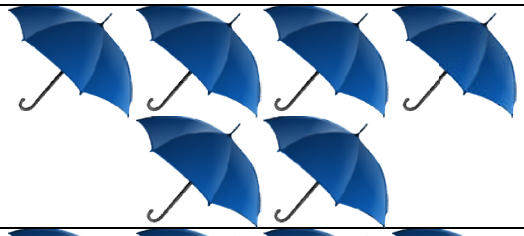
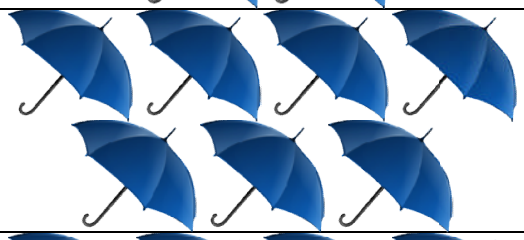
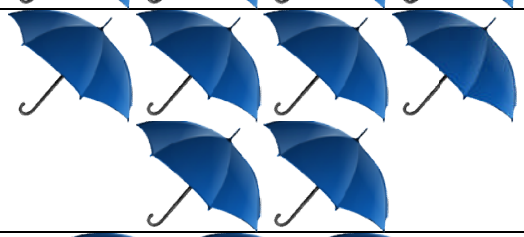
Only items that have votes or numbers are recorded; any items that could have been voted for but weren't are not shown, this can be misleading.

6) Draw a picture graph that shows the number of days it rained in a series of weeks from the table of data. Make up your own symbol and scale

WEEK NUMBER	NUMBER OF RAINY DAYS
1	2
2	4
3	0
4	6
5	7
6	4
7	5
8	3
9	2
10	0

 = 1 rainy day

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WEEK NUMBER	NUMBER OF RAINY DAYS
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

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7) What do you think the following picture graph is showing? (Hint: It is not showing size)

MY FAMILY

GRANDAD	
GRANDMA	
DAD	
MUM	

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ME	
BROTHER	
BABY SISTER	
PET DOG	

The clues are that people who are married to each other appear as a similar “size”, and that the dog is “larger” than his sister and about the same “size” as his brother. The picture graph is comparing ages of people in his family; the larger the image, the older the person (or dog)

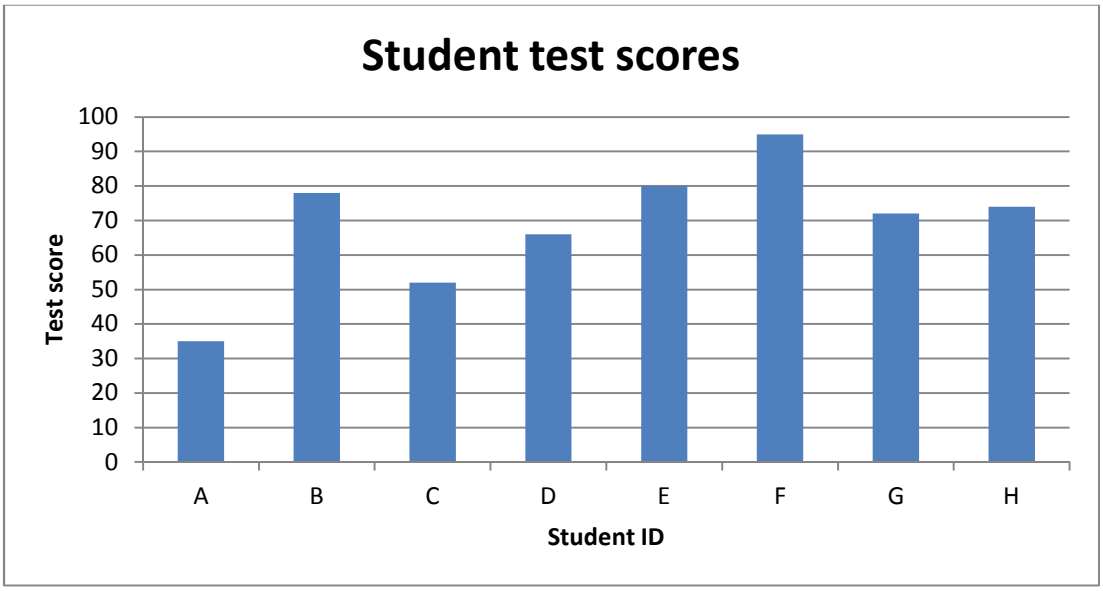
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Exercise 3

Column Graphs

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1) The following graph shows the test scores for a group of students



a) Which student scored the highest and what was their score?

Student F scored approximately 94

b) How many students failed the test?

Student A was the only student who failed the test (under 50)

c) One student only just passed. What was their mark?

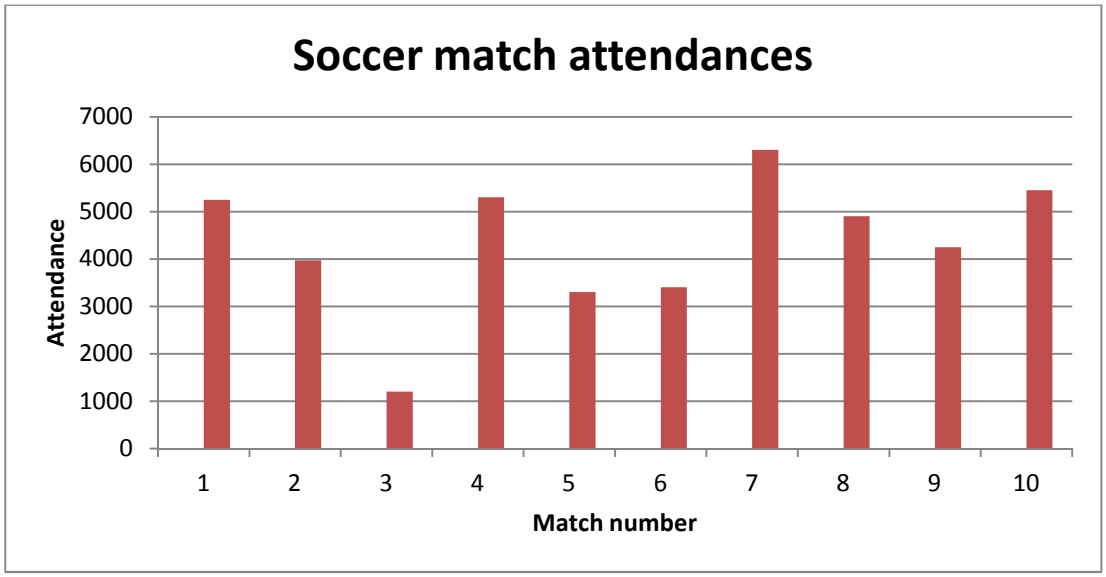
Student C scored just over 50

d) Name two students whose marks were almost the same

B and E, or G and H

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2) The attendances at the soccer matches from exercise 2, question 1 are shown in the column graph below



a) Estimate the attendance for game 1 and compare it with the estimate of the attendance using the picture graph from exercise 2

52,000

b) Repeat for game 10

54,000

c) What game had the highest attendance and approximately what was that attendance?

Game 7, 62,000

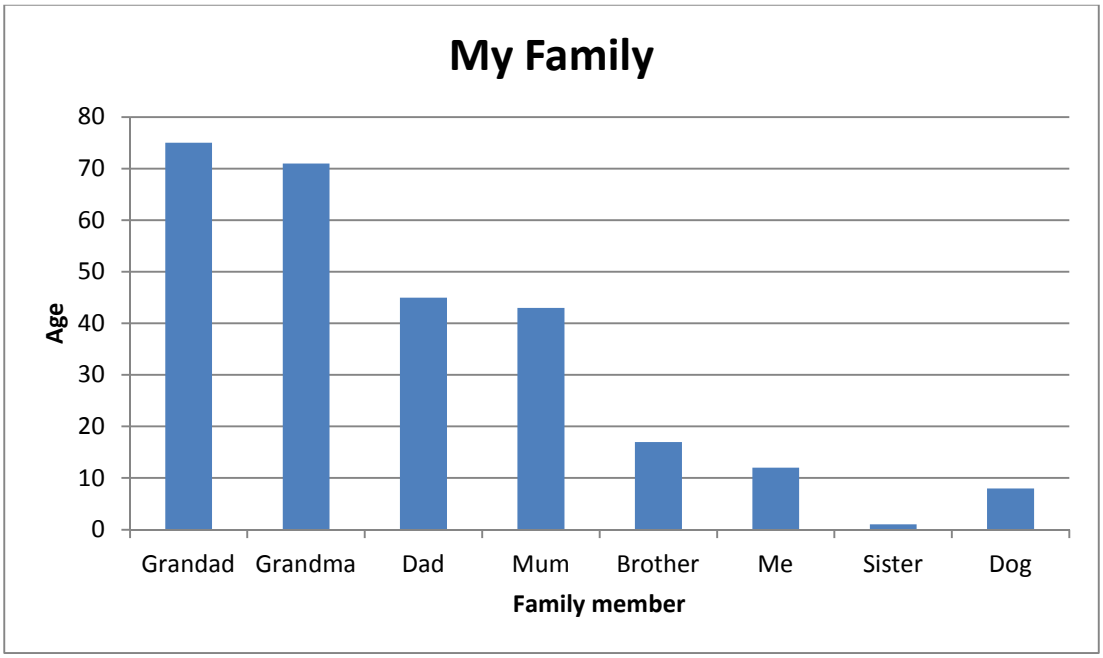
d) From your answers state an advantage of using column graphs over picture graphs

Figures can be represented more accurately and don't have to be rounded to suit the value of the picture used

The scale can be shown more accurately

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3) The following graph shows the ages of the members of a student's family



a) Who is the oldest in the family and how old are they?

Grandad, approximately 75

b) Who is the youngest and how old are they?

Sister, approximately 1

c) Approximately how old is the dog?

8

d) How much older is the student's dad than the student?

45 – 11 = 34 years older

e) From this question and the corresponding question in exercise2, discuss an advantage and a disadvantage of using column graphs to represent data

Advantage: Numbers such as 11 can be represented in column graphs, whereas in picture graphs such an age may be hard to represent accurately (if say each item represented 10 years)

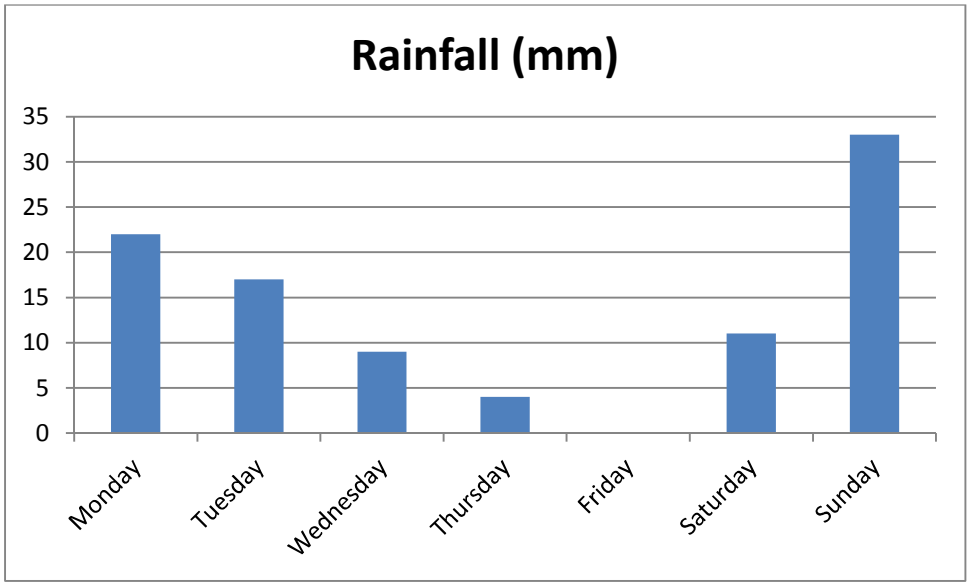
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Disadvantage: Exact values can be hard to read (e.g. 34)

4) Draw a column graph that represents the following data

Rainfall figures for week in mm

Day	Rainfall (mm)
Monday	22
Tuesday	17
Wednesday	9
Thursday	4
Friday	0
Saturday	11
Sunday	33

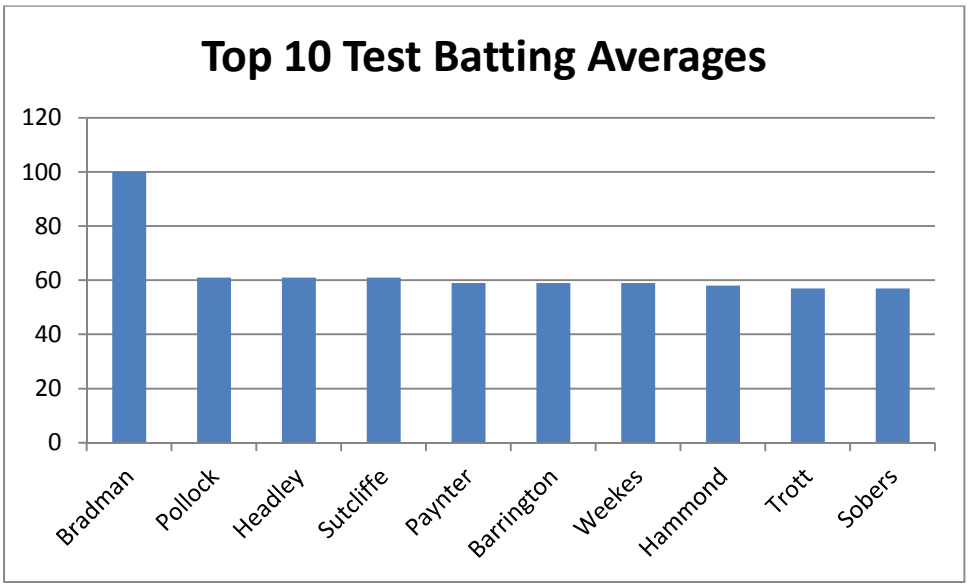


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5) The following table shows the ten best test batting averages of all time (rounded to the nearest run)

Name	Average
Bradman	100
Pollock	61
Headley	61
Sutcliffe	61
Paynter	59
Barrington	59
Weekes	59
Hammond	58
Trott	57
Sobers	57

Draw a column graph to represent the above data, and by comparing the data for Bradman to the others, discuss one advantage and one disadvantage of using column graphs to represent such a data set



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Column graphs are useful for showing values that stand out from the rest (called outliers), however it can be hard to tell the difference if a lot of the values are nearly the same

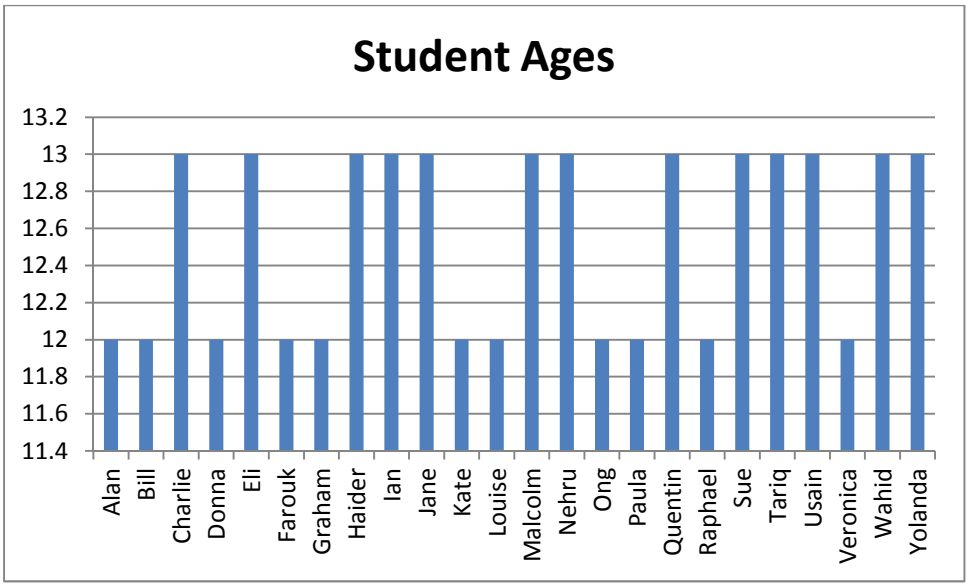
6) The teacher of a large year group wishes to plot the ages of her students on a graph. Their names and ages are shown in the table below

Name	Age
Alan	12
Bill	12
Charlie	13
Donna	12
Eli	13
Farouk	12
Graham	12
Haider	13
Ian	13
Jane	13
Kate	12
Louise	12
Malcolm	13
Nehru	13
Ong	12
Paula	12
Quentin	13
Raphael	12
Sue	13
Tariq	13
Usain	13

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Veronica	12
Wahid	13
Yolanda	13

a) Plot the data on a column graph.



b) Imagine we had to graph the ages of year 7 students in the whole state. Using your graph as a guide, explain why a column graph is not suitable for displaying this data. Can you think of a better alternative?

The graph would be far too large; it would stretch for probably a kilometre!

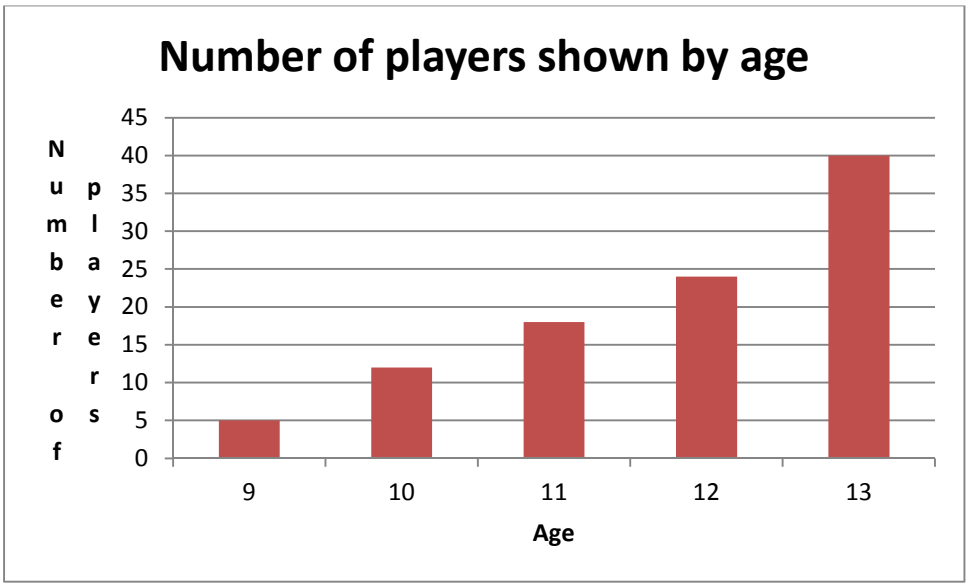
A possible better alternative would be to have one column that shows all students of each certain age

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7) A football club wanted to graphically show the ages of all players in their under 14 teams. Firstly they counted all the ages of the players and totalled the number of players of each age.

Age	Number of players
9	5
10	12
11	18
12	24
13	40

a) Draw this data as a column graph, and compare it to the column graph of question 6.



b) Which way of showing the players' ages graphically is easier to draw and shows the data in a smaller easier to read graph?

This graph since it is a lot smaller than the alternative

c) What is a disadvantage of graphing the ages in this way?

You cannot see the individual names as it shows totals only

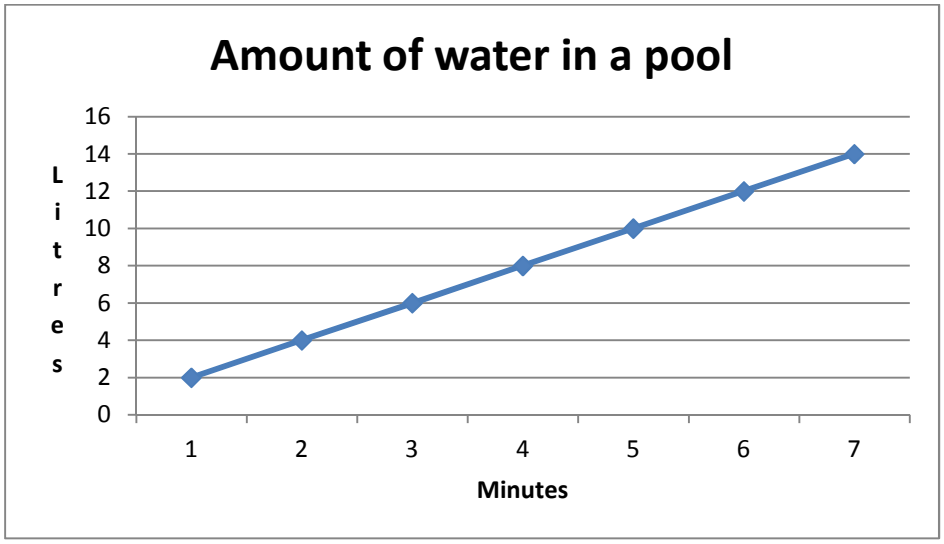
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Exercise 4

Line Graphs

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1) A pool is being filled with a hose. The graph below shows the number of litres in the pool after a certain number of minutes



a) How much water was in the pool after 3 minutes?

The dot in line with 3 minutes shows 6 litres

b) How many minutes did it take to put 12 litres into the pool?

The dot in line with 12 litres shows 6 minutes

c) How fast is the pool filling up?

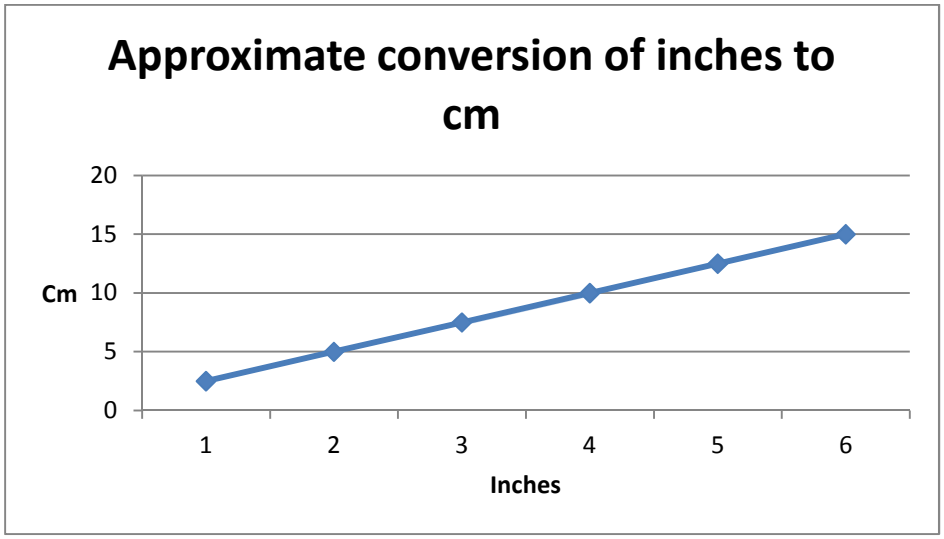
The gap between the dots shows one minute and 2 litres; so every minute, 2 litres of water goes into the pool

d) How many litres will be in the pool after 8 minutes, assuming it keeps getting filled at the same rate?

8 x 2 = 16 litres

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2) The graph below shows approximately how many cm are equal to a certain number of inches



a) Approximately how many cm are there in 4 inches?

The dot in line with 4 inches shows 10 cm

b) Approximately how many inches are there in 5 cm?

The dot in line with 5 cm shows 2 inches

c) About how many cm equal one inch?

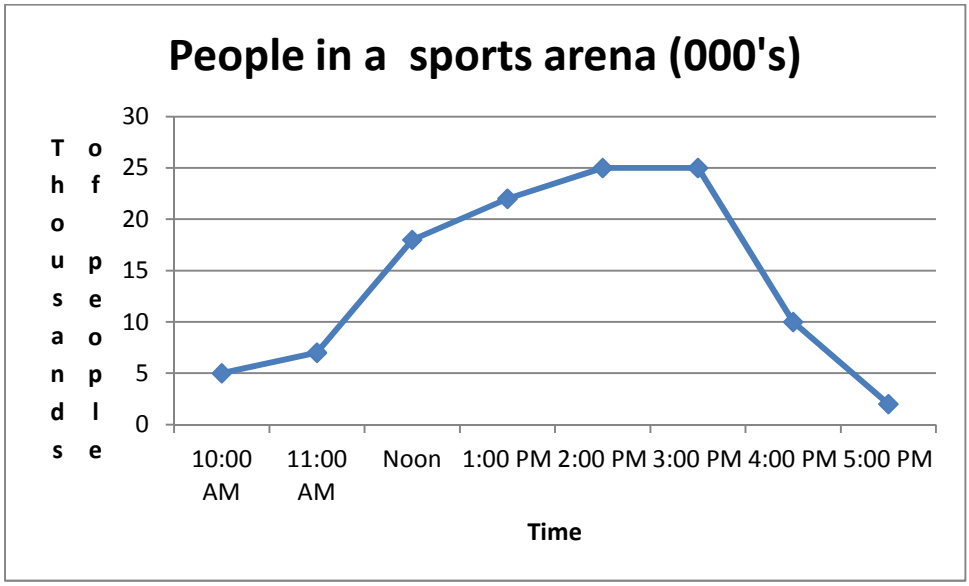
For every inch, the cm rises by 2.5; therefore there are approximately 2.5 cm in 1 inch

d) Approximately how many cm are in 8 inches?

8 x 2.5 = 20 cm

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3) The graph below shows how many people were at a sports arena at various times of the day



a) How many people were in the ground at 11 AM?

The dot in line with 11 am shows about 6000 people

b) When were there approximately 10,000 people in the ground?

10,000 people is in line with the dot for 4:00 pm

c) At what time would the game have started? Explain your answer

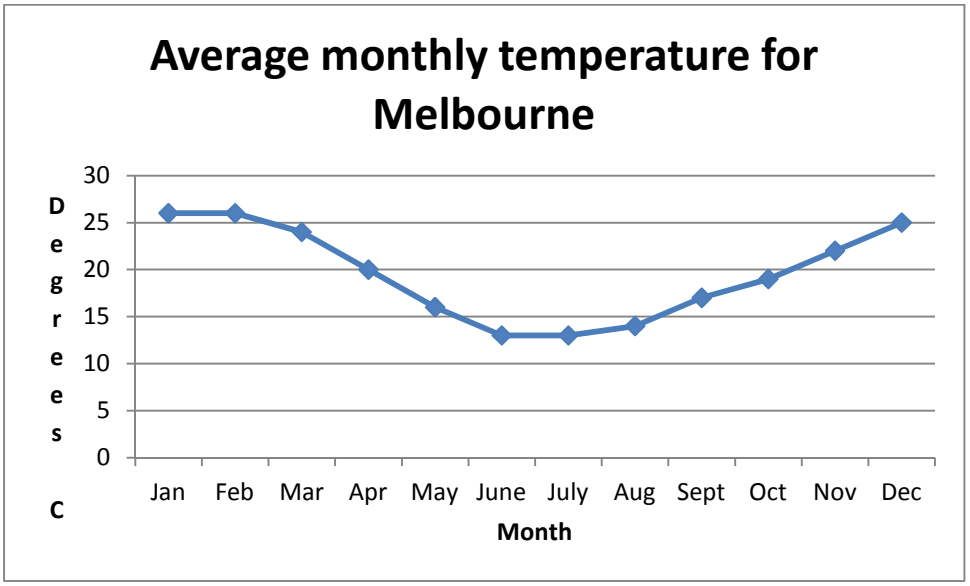
Probably 2:00 pm since the maximum crowd was there and no one else came in after that time

d) Why can't you say that the number of people in the ground at 3:30 PM was 15,000?

There is no data point (dot) to actually say that that amount was present at that time; the line is simply joining the two data points

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4) The graph below shows the average daily temperature per month for Melbourne



a) What is the average daily temperature in December?

25 degrees

b) Which months are the coldest?

June & July

c) Name two non consecutive months when the average temperatures are the same

There are none exactly, but March & December are close

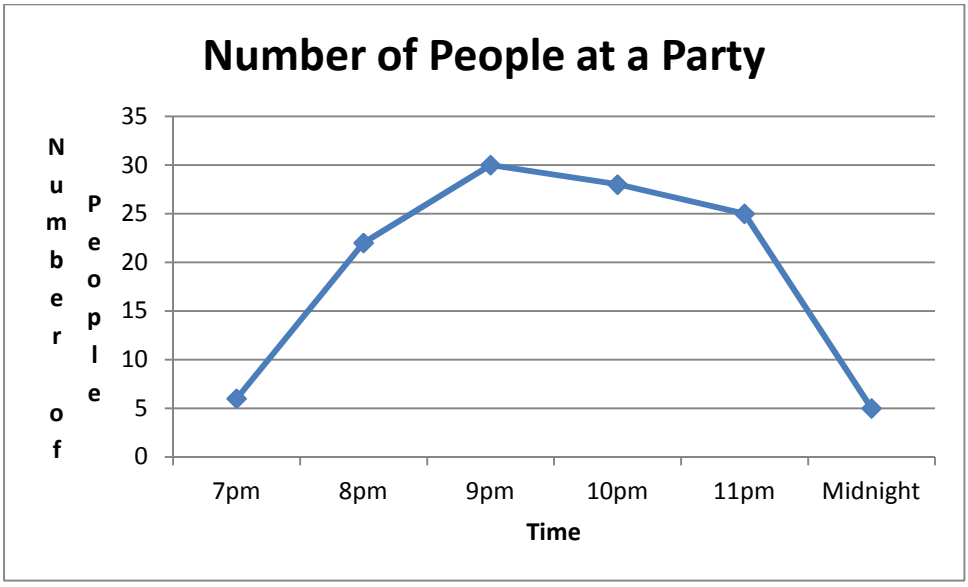
d) Does the graph show that temperatures in Melbourne will never go above 26 degrees? Explain your answer

No; these show average temperatures, some temperatures will be above the average and some below

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5) Graph the following data in a line graph

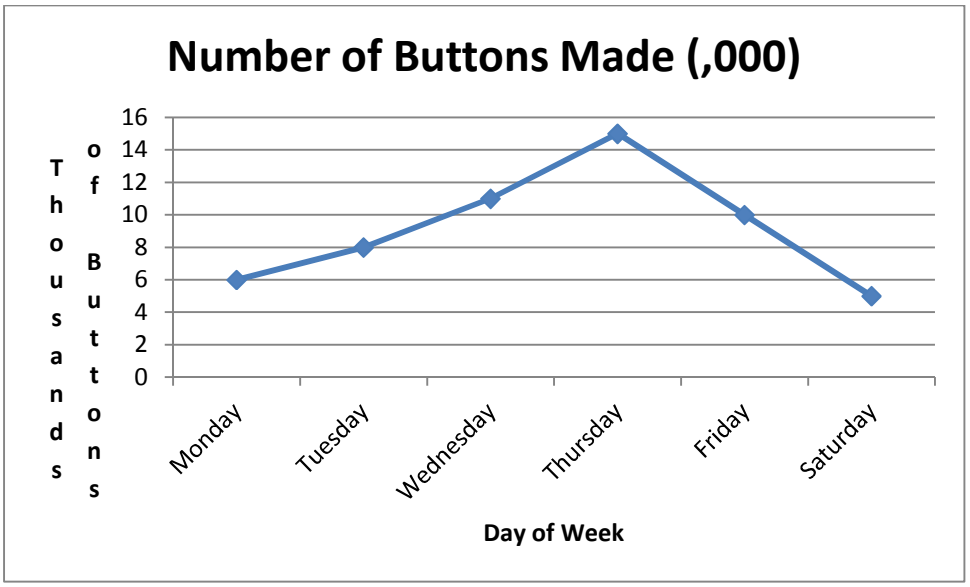
Time	Number of people at a party
7 PM	6
8 PM	22
9 PM	30
10 PM	28
11 PM	25
Midnight	5



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6) Graph the following data in a line graph (Consider your scale)

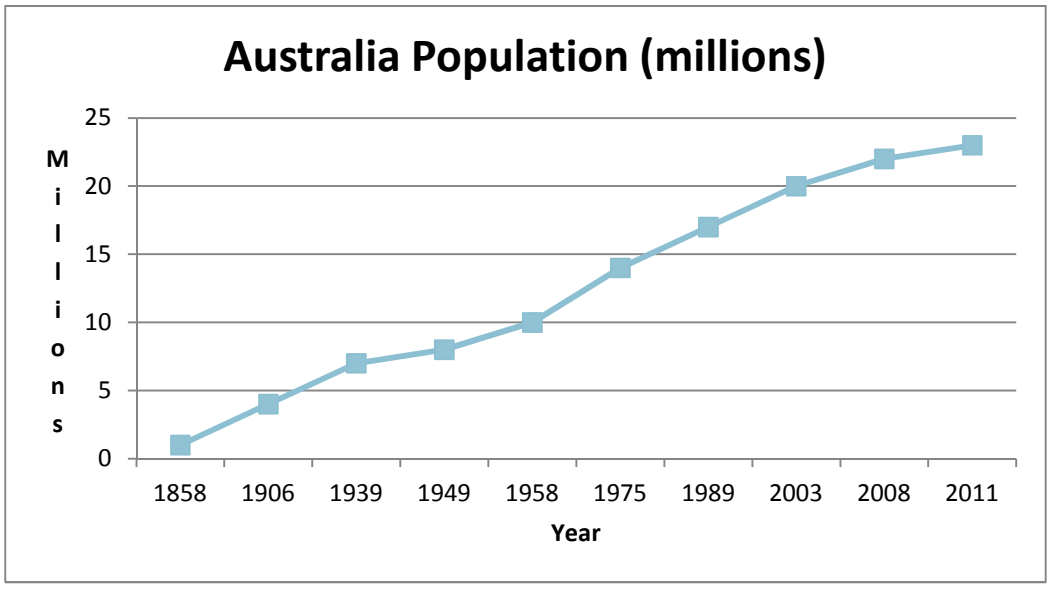
Day	Number of buttons made at factory (thousands)
Monday	6
Tuesday	8
Wednesday	11
Thursday	15
Friday	10
Saturday	5



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7) Graph the following data that shows the population of Australia over time

Year	Population (approximate)
1858	1 million
1906	4 million
1939	7 million
1949	8 million
1958	10 million
1975	14 million
1989	17 million
2003	20 million
2008	22 million
2011	23 million



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Year 5 Mathematics

Algebra & Patterns

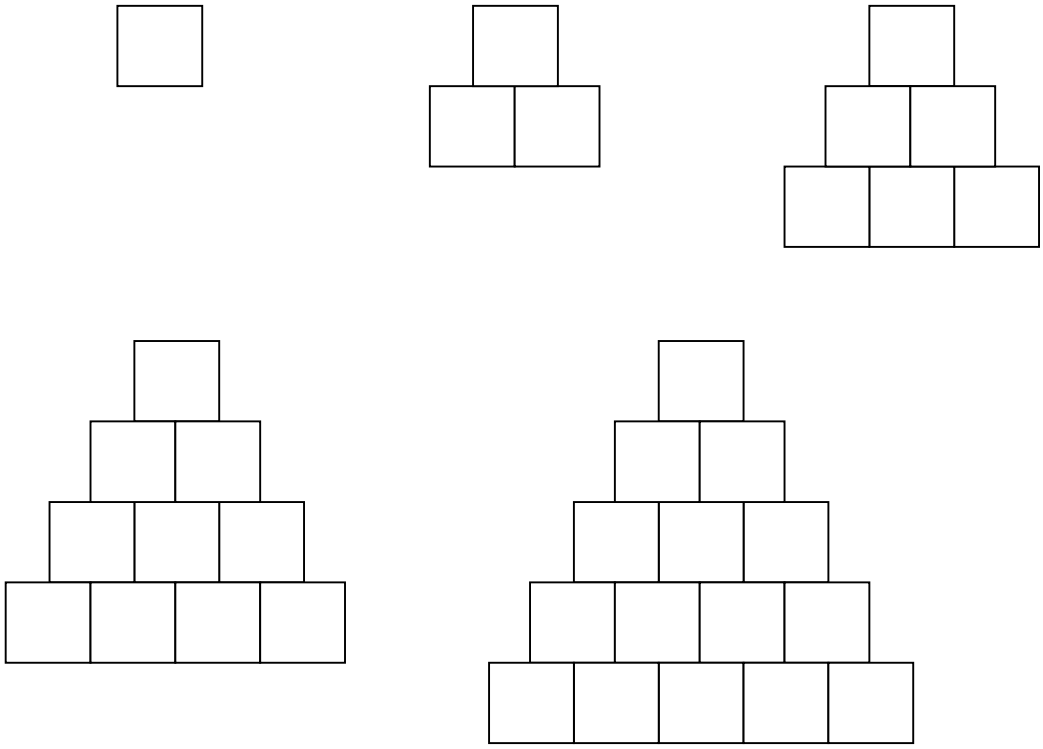
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Exercise 1

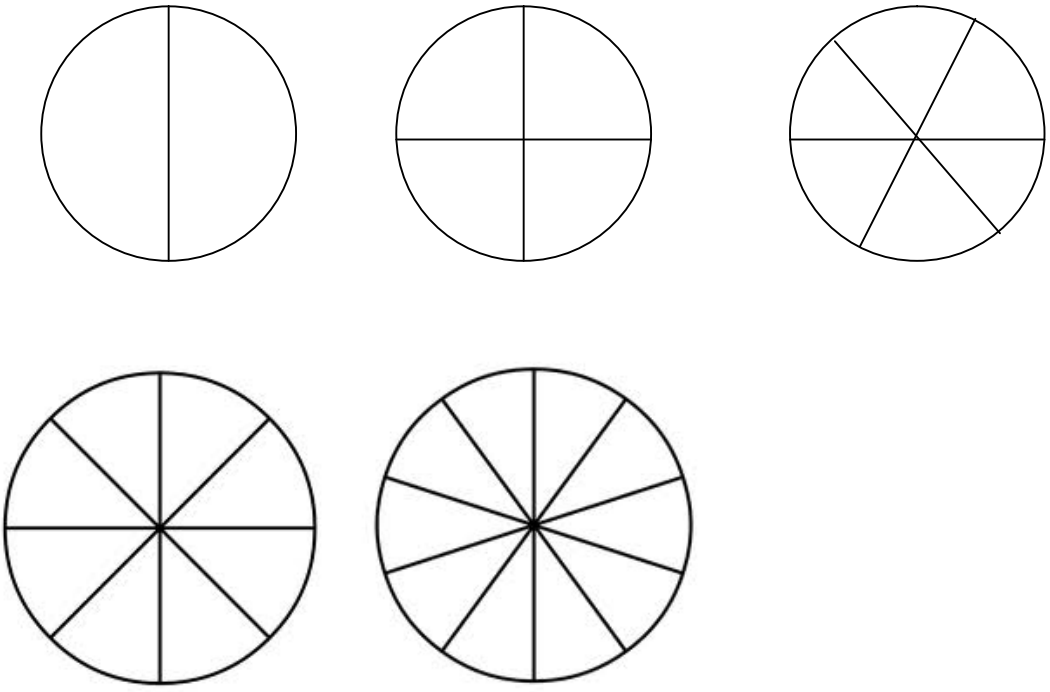
Simple Geometric Patterns

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1) Draw the next two diagrams in this series

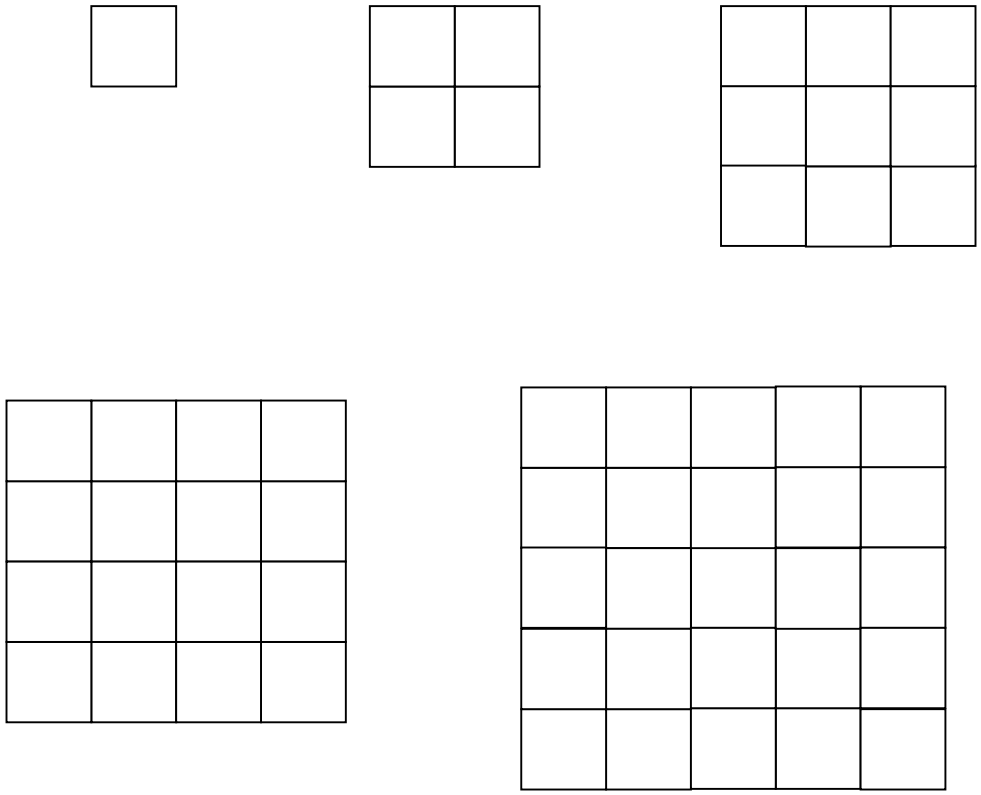


2) Draw the next two diagrams in this series

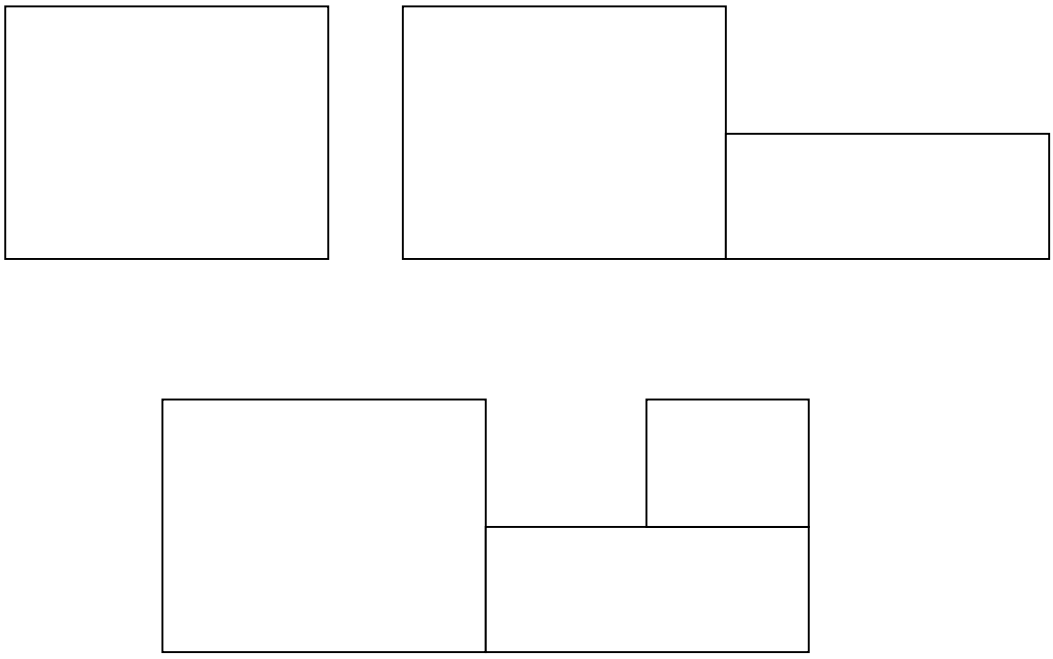


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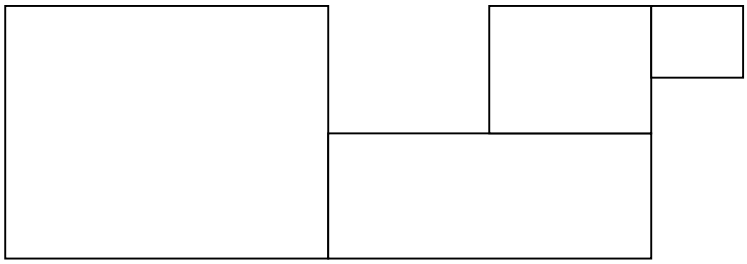
3) Draw the next two diagrams in this series

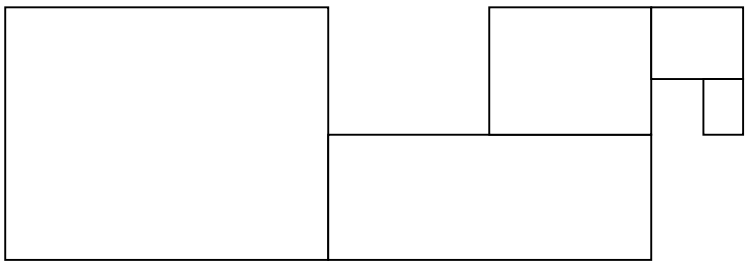


4) Draw the next two diagrams in this series

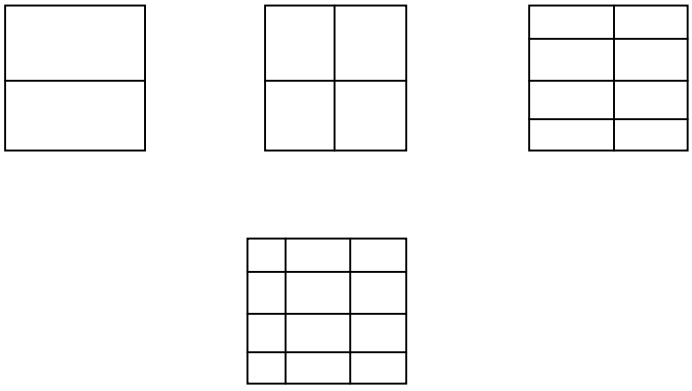


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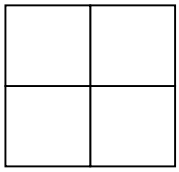


5) To make two equal pieces of chocolate from a square block one cut is required. To make four equal pieces two cuts are required. How many cuts are needed to make 8 equal pieces? How many cuts are required to make 12 equal pieces?



4 cuts for 8 pieces, 5 cuts for 12 pieces

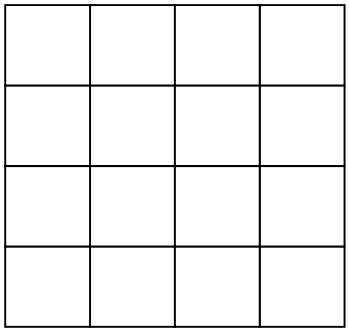
6) There are 5 squares on a 2 x 2 chessboard



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Four small squares and one large square

How many squares on a 4 x 4 chessboard?



There is 1 large square (4 x 4)

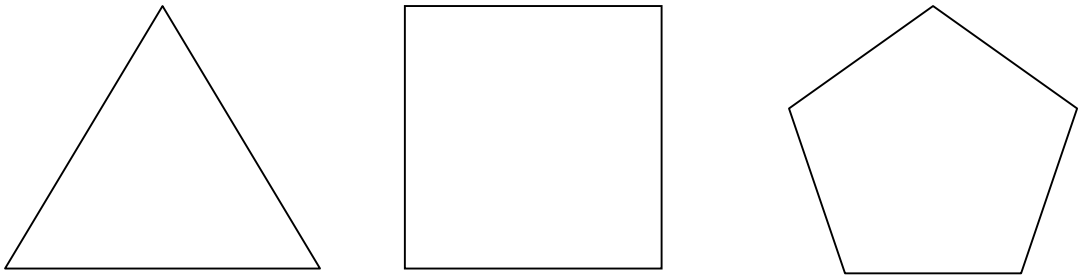
There are 4 smaller squares (3 x 3)

There are 9 smaller squares (2x2)

There are 16 smallest squares (1x1)

There are 30 squares on a 4 x 4 chessboard

7) Measure and add up the internal angles of the following shapes



Use you results to predict the sum of the internal angles of a hexagon (6 sides) and a heptagon (7 sides)

Equilateral Triangle (3 angles x 60°) = 180°

Square (4 angles x 90°) = 360

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Regular Pentagon (5 angles x 108°) = 540°

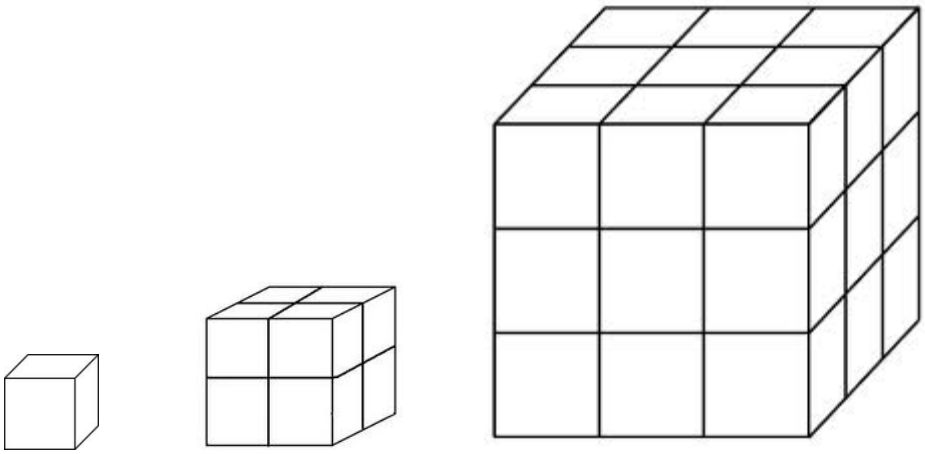
Every time a side is added, the sum of the angles increases by 180°

Therefore Hexagon = 720°

Heptagon = 900°

Extension: Can you calculate the size of each angle in the above shapes?

8) How many cubes in the next two shapes in this series?



1 cube = 1 x 1 x 1 8 cubes = 2 x 2 x 2 27 cubes = 3 x 3 x 3

The number of cubes is equal to the length of one side cubed

A cube of length 4 units would have 4 x 4 x 4 = 64 small cubes

A cube of length 5 units would have 5 x 5 x 5 = 125 small cubes

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Exercise 2

Simple Number Patterns

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1) For the following series, fill in the next two terms

a) 1, 3, 5, 7

Each term is 2 more than the previous

1, 3, 5, 7, 9, 11

b) 2, 4, 8, 16

Each term is double the previous

2, 4, 8, 16, 32, 64

c) 1, 4, 9, 16

Each term is the square of the counting numbers in order.

OR

Add 3 to the first term, 5 to the second term, 7 to the third term etc

1, 4, 9, 16, 25, 36

d) 1, 3, 6, 10

Add 2 to the first term, 3 to the second, 4 to the third etc

1, 3, 6, 10, 15, 21

2) For the following series, fill in the next two terms

a) 5, 10, 15, 20

These are the multiples of 5

5, 10, 15, 20, 25, 30

b) 32, 16, 8, 4

Halve the previous number

32, 16, 8, 4, 2, 1

c) 100, 90, 80, 70

Subtract 10 from the previous number

100, 90, 80, 70, 60, 50

d) 64, 49, 36, 25

The square of 8, the square of 7, the square of 6, the square of 5 etc

OR

Subtract 15, subtract 13, subtract 11 etc

64, 49, 36, 25, 16, 9

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3) Fill in the blanks in the following

a) 2, 6, ___, 14, 18, ___

Add 4 to each number to get the next

2, 6, 10, 14, 18, 22

b) ___, 22, 33, ___, 55

Add 11 to each number to get the next

11, 22, 33, 44, 55

c) 1, 3, ___, 27, ___, 243

Multiply each number by 3 to get the next number

1, 3, 9, 27, 81

d) 0.5, 1, 1.5, ___, ___

Add 0.5 to each number to get the next

0.5, 1, 1.5, 2, 2.5

e) $\frac{1}{2}, \frac{1}{4}$, ___, $\frac{1}{16}$, ___

Halve each fraction to get the next

$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}$

4) What are the next three numbers of the following series?

0, 1, 1, 2, 3, 5, 8

Each number is the sum of the previous two numbers

0, 1, 1, 2, 3, 5, 8, 13, 21

This is a famous sequence called the Fibonacci sequence

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5) Thomas walked 3km on Monday, 6km on Tuesday, and 9km on Wednesday. If this pattern continues

a) How far will he walk on Friday?

Each day he walks 3 km more than the previous day.

On Thursday he will walk 12 km

On Friday he will walk 15 km

b) What will be the total distance he has walked by Saturday?

3 + 6 + 9 + 12 + 15 = 45 km

6) At the start of his diet, a man weighs 110kg. Each week he loses 4kg.

a) How much weight will he have lost by the end of week 3?

3 x 4 kg = 12 kg

b) How much will he weigh by the end of week 4?

Will have lost 4 x 4 kg = 16 kg

He will weigh 110 – 16 = 94 kg

7) A pond of water evaporates at such a rate that at the end of each day there is half as much water in it than there was at the start of the day. If there was 128 litres of water in the pond on day one, at the end of which day will there be only 8 litres of water left?

At the end of day one there will be 64 litres left
At the end of day 2 there will be 32 litres left
At the end of day 3 there will be 16 litres left
At the end of day 4 there will be 8 litres left

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8) Fill the blanks in the following series

a) 40, 42, 39, 43, 38, 44, ___, ____

Add 2, subtract 3, add 4, subtract 5, add 6: ...

37, 45

b) 100, 200, 50, 100, 25, ___, ___

Double, quarter, double, quarter: ....

50, 12.5

c) 1, ___, 10, 16, 23, ___

The middle three terms are found by adding 6 then 7 to the previous number.

The pattern needs a number that gives 10 when 5 is added to it, but also results from adding 4 to 1

Blanks are 5, 31

d) 1, 2, 5, 26, ___, ___

Since the numbers increase quickly, squaring of numbers is probably involved. Squaring the previous number does not work, but adding 1 to the result does

Blanks are 677, 458330

9) Complete the following series

a) 8, 12, 18, 27, ___

Each number is 1.5 times the previous number

40.5, 60.75

b) 4, 6, 10, 18, 34, ___, ___

Add 2, add 4, add 8, add 16; the next two additions will be 32 and 64, and the numbers are:

66, 130

c) 100, 60, 40, 30, ___, ___

Subtract 40, subtract 20, subtract 10; the next two subtractions will be 5 and 2.5, and the numbers are:

25, 22.5

d) 7.5, 7, 8.5, ___, 9.5, ___

Subtract 0.5, add 1.5; see if repeating will make pattern

Blanks will be 8, 9

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10) A bug is crawling up a wall. He crawls 2 metres every hour, but slips back one metre at the end of each hour from tiredness.

a) How far up the wall will he be in 5 hours?

Hour	Distance up wall at beginning of hour	Distance up wall before he slips back	Distance up wall after he slips back
1	0	2	1
2	1	3	2
3	2	4	3
4	3	5	4
5	4	6	5

Form the table, after 5 hours he will be 5 metres up the wall

b) How long will it take him to reach the top of a 10 meter wall?

The answer would seem to be 10 hours, but if you continue the table.....

Hour	Distance up wall at beginning of hour	Distance up wall before he slips back	Distance up wall after he slips back
1	0	2	1
2	1	3	2
3	2	4	3
4	3	5	4
5	4	6	5
6	5	7	6
7	6	8	7
8	7	9	8
9	8	10	X

He will not slip back once he reaches the top of the wall, therefore it will take him 9 hours to reach the top

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Exercise 3

Rules of patterns & Predicting

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Different bacteria have different reproduction and death rates, so a group of different bacteria samples will have different populations depending on what type they are.

The populations of different types of bacteria were measured at one minute intervals, and the numbers present were recorded in separate tables which are shown in questions 1 to 7.

For each question you are required to:

- Fill in the missing figure
- Work out a rule that relates the number of minutes passed to the number of bacteria in the sample
- Use this rule to predict the number of bacteria in the sample after 100 minutes

The following example will help you

Minutes	1	2	3	4	10
Number	2	4	6	8	

It can be seen that the population increases by 2 bacteria every minute. Therefore in six minutes (the amount of time between 4 and 10), the population will increase by 12 bacteria (6 x 2). Therefore the population after 10 minutes will be 8 + 12 = 20 bacteria

To predict the population for longer time periods it is useful to find a rule that relates the number of minutes to the number of bacteria and apply that rule.

After 1 minute the population was 2 bacteria. This would suggest that if you add 1 to the number of minutes you will get the number of bacteria. The rule must work for every number of minutes. If you take 2 minutes and add 1 to it you get 3 bacteria, which does not match the table, therefore the rule is wrong

Another rule may be that you multiply the number of minutes by 2 to get the number of bacteria. This certainly works for 1 minute. What about 2 minutes or 3 minutes? If you multiply any of the minutes by 2 you will get the number of bacteria. Therefore you have found the rule.

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The rule should be stated:

The number of bacteria can be found by multiplying the number of minutes by 2

Use the rule to check your answer for 10 minutes found earlier (10 x 2 = 20, therefore correct), and to predict the number of bacteria after 100 minutes (100 x 2 =200)

NOTE: Some of the rules will involve a combination of multiplication and addition, or multiplication and subtraction

1)

Minutes	1	2	3	4	10
Number	4	5	6	7	

Adding 3 to the number of minutes works for the first minute, what about the others?

2 + 3 = 5, 3 + 3 = 6, 4 + 3 = 7.

The rule is

The number of bacteria can be found by adding 3 to the number of minutes.

The missing entry is 13 bacteria.

The number of bacteria after 100 minutes is 100 + 3 = 103

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2)

Minutes	1	2	3	4	10
Number	3	5	7	9	

Adding 2 to the number of minutes works for the first minute, but not for the others.
Tripling the number of minutes works for the first minute, but not for the others.
Therefore the rule must be a combination of multiplication/division and addition/subtraction

If you double the number of minutes and add 1 to the result, this works for the first minute; what about the others?

(2 x 2) + 1 = 5
(3 x 2) + 1 = 7
(4 x 2) + 1= 9

The rule is:

The number of bacteria can be found by multiplying the number of minutes by 2 and adding 1 to the result

The missing entry is (10 x 2) + 1 = 21 bacteria

After 100 minutes there are (100 x 2) + 1 = 201 bacteria

3)

Minutes	1	2	3	4	10
Number	10	20	30	40	

Multiplying the number of minutes by 10 works for the first minute, also 2 x 10 = 20, 3 x 10 = 30, 4 x 10 = 40

The rule is:

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The number of bacteria can be found by multiplying the number of minutes by 10

The missing entry is 100 bacteria

After 100 minutes there are 100 x 10 = 1000 bacteria

4)

Minutes	1	2	3	4	10
Number	2	5	8	11	

Doubling the number of minutes works for the first minute, but not for any others

Adding 1 to the number of minutes works for the first minute, but not for any others

Therefore the rule must be a combination

Doubling and adding or doubling and subtracting does not work

Tripling the number of minutes and subtracting 1 works for the first minute; also:

(2 x 3) – 1 = 5
(3 x 3) – 1 = 8
(4 x 3) – 1 = 11

The rule is

The number of bacteria can be found by multiplying the number of minutes by 3 and subtracting 1

The missing entry is (10 x 3) -1 = 29 bacteria

After 100 minutes there are (100 x 3) – 1 = 299 bacteria

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5)

Minutes	1	2	3	4	10
Number	1	3	5	7	

Setting the number of bacteria equal to the number of minutes works for the first minute, but not the others

The rule must be a combination

Doubling the amount of minutes and subtracting 1 works for the first minute; also:

(2 x 2) - 1 = 3
(3 x 2) - 1 = 5
(4 x 2) - 1 = 7

The rule is:

The number of bacteria can be found by multiplying the number of minutes by 2 and subtracting 1

The missing entry is (10 x 2) -1 = 19 bacteria

After 100 minutes there are (100 x 2) -1 = 199 bacteria

6)

Minutes	1	2	3	4	10
Number	4	6	8	10	

If you multiply the number of minutes by 4 this works for the first minute, but not the others

If you add 3 to the number of minutes this works for the first minute but not the others.

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Therefore the rule involves a combination

If you multiply the number of minutes by 2, then add 2 this works for the first minute; also:

(2 x 2) + 2 = 6
(3 x 2) + 2 = 8
(4 x 2) + 2 = 10

The rule is

The number of bacteria can be found by multiplying the number of minutes by 2 then adding 2

The missing entry is (10 x 2) + 2 = 22 bacteria

After 100 minutes there are (100 x 2) + 2 = 202 bacteria

7)

Minutes	1	2	3	4	10
Number	110	120	130	140	

If you multiply the number of minutes by 100 this works for the first minute, but not the others.

If you add 109 to the number of minutes this works for the first minute but not the others

Therefore the rule involves a combination

Every quantity is in the hundreds so a good guess would be that 100 is added to something.

If we multiply the number of minutes by 10 then add 100 this works for the first minute; also:

(2 x 10) + 100 = 120

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(3 x 10) + 100 = 130

(4 x 10) + 100 = 140

The rule is

The number of bacteria can be found by multiplying the number of minutes by 10, then adding 100

The missing entry is (10 x 10) + 100 = 200 bacteria

After 100 minutes there are (100 x 10) + 100 = 1100 bacteria

8) The time for roasting a piece of meat depends on the weight of the piece being cooked. The directions state that you should cook the meat for 30 minutes at 260 degrees, plus an extra 10 minutes at 200 degrees for every 500 grams of meat How long would the following pieces of meat take to cook?

a) 500 grams of meat

30 minutes plus one lot of 10 minutes (for 500 grams) = 40 minutes

b) 1 kg

30 minutes plus two lots of 10 minutes (1 kg = 2 x 500 g) = 50 minutes

c) 2 kg

30 minutes plus four lots of 10 minutes (2 kg = 4 x 500 g) = 70 minutes

d) 3.5 kg

30 minutes plus seven lots of 10 minutes (3.5 kg = 7 x 500 g) = 100 minutes

9) Taxis charge a flat charge plus a certain number of cents per kilometre. A man took a taxi ride and noted the fare at certain distances

- After 1 km the fare was $2.50
- After 3 km the fare was $3.50
- After 10 km the fare was $7.00

What was the flat charge, and how much did each kilometre cost?

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For the 2 km from 1 to 3 km the charge went up by $1

Therefore each km costs 50c

If 1 km costs 50c then the charge at zero km (the flat charge) must be $2.50 – 50c = $2

Therefore the charges are $2 flat fee plus 50c per km

Check the answer by calculating the charge at 10 km

Charge = (10 x 50c) + $2 = $7, which is correct

10) A business wanted to get two quotes to fix their truck, so they approached two different mechanics, Alan and Bob. Their quotes were:

Alan: $100 call out fee plus $40 per hour

Bob: $200 call out fee plus $20 per hour

Which mechanic should the company hire?

The answer is that it depends on how long the job will take, and is best shown in a table

Number of hours	**1**	**2**	**3**	**4**	**5**	**6**
Cost of Alan	140	180	220	260	300	340
Cost of Bob	220	240	260	280	300	320

For any jobs less than 5 hours, Alan is cheaper, whilst Bob should be hired for any jobs over 5 hours; at 5 hours their costs are equal

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Year 5 Mathematics

Measurement:
Length & Area

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Exercise 1

Units of Measurement

Converting & Applying

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1) Convert the following to metres

a) 3245 mm

3.245 m

b) 809 cm

8.09 m

c) 32 km

32,000 m

d) 5.43 km

5430 m

e) 70 cm

0.7 m

2) Convert the following to centimetres

a) 41.4 m

4140 cm

b) 1762 mm

176.2 cm

c) 4 m

400 cm

d) 0.8 km

800 m

e) 9 mm

0.9 cm

3) Convert the following to millimetres

a) 9 cm

90 mm

b) 0.3 m

300 mm

c) 1.27 m

1270 mm

d) 4 km

4,000,000 mm

e) 19.2 m

19200 mm

4) Convert the following to square centimetres

a) 10 square metres

100,000 cm^2

b) 100 square millimetres

1 cm^2

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c) 0.4 square kilometres

4,000,000,000 cm^2

d) 0.142 square metres

1420 cm^2

e) 3174 square millimetres

31.74 cm

5) Which is larger?

a) 145 mm or 1.45 cm

1.45 cm = 14.5 mm
< 145 mm

b) 73 km or 7300 m

73 km = 73,000 m > 7300 m

c) 193 cm or 1930 mm

193 cm = 1930 mm

d) 10.3 m or 1030 mm

10.3 m = 10,300 mm
>1030 mm

e) 0.5 km or 5000 cm

0.5 km = 50,000 cm
> 5000 cm

6) Which is smaller?

a) 144 square mm or 1.44 square cm

144 mm^2 = 1.44 cm^2

b) 1 square km or 100000 square metres

1 km^2 = 1,000,000 m^2
>100,000 m^2

c) 178 square cm or 0.178 square metres

178 cm^2 = 0.0178 cm^2
< 0.178 m^2

d) 100 square metres or 1000 square centimetres

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7) Each day for four days, Bill walks 2135 metres. Ben walks 1.2 km on each of five days. Who has walked the furthest?

Bill walks 4 x 2135 m = 8540 m = 8.54 km

Ben walks 5 x 1.2 km = 6 km

Bill walks further

8) Mark has to paint a floor that has an area of 180 square metres, whilst Tan has to paint a floor that has an area of 180000 square centimetres. Who will use more paint?

180000 cm^2 = 18 m^2

Mark uses more paint

9) A snail travels 112 cm in 10 minutes, whilst a slug takes 20 minutes to go 22.4 metres. Which creature would cover more ground in an hour and by how much?

In one hour (6 x 10 minutes) the snail travels 6 x 112 cm = 672 cm = 6.72 m

In one hour (3 x20 minutes) the slug travels 3 x 22.4 m = 67.2 m

The slug travels 67.2 – 6.72 = 60.48 m more in one hour

10) Alan walks 1.4 km to the end of a long road, then he walks another 825 metres to the next corner. He then walks 5 metres to the front of a shop and goes through the entrance which is 600 cm. How far has he walked altogether? Give your answer in km, m, and cm

In meters he walked 1400 + 825 + 5 + 6 = 2236 m

2236 m = 2.236 km

2236 m = 223,600 cm

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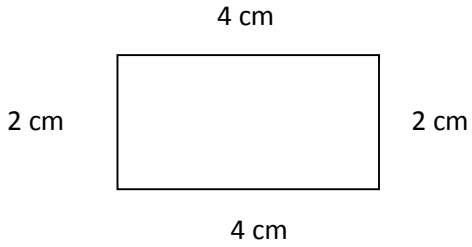
Exercise 2

Simple Perimeter Problems

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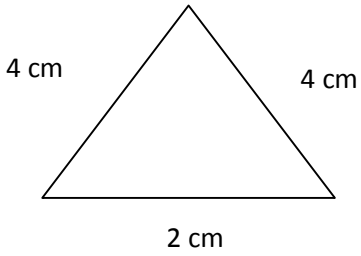
1) Calculate the perimeter of the following

a)



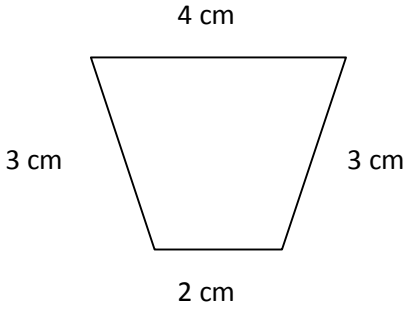
4 + 2 + 4 + 2 = 12 cm

b)



4 + 4 + 2 = 10 cm

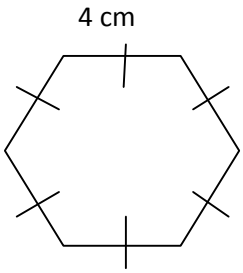
c)



4 + 3 + 2 + 3 = 12 cm

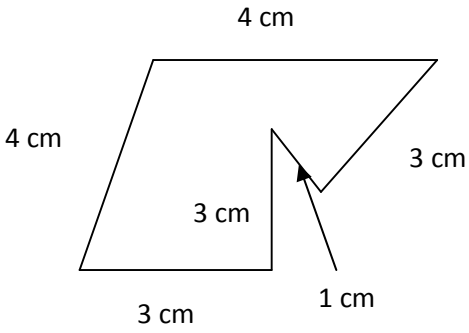
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d)



6 x 4cm = 24 cm

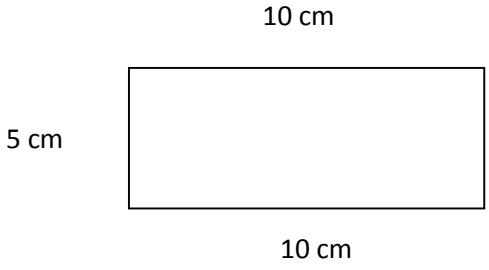
e) A



4 + 4 + 3 + 1 + 3 + 3 = 18 cm

2) The perimeter of the following shapes is 30 cm. Calculate the unknown side length(s)

a)

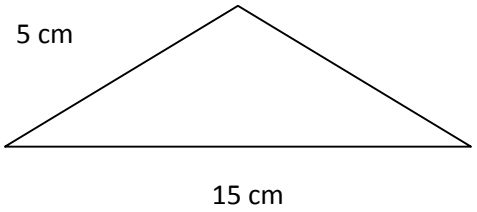


5 + 10 + 10 =25

30 – 25 = 5 cm

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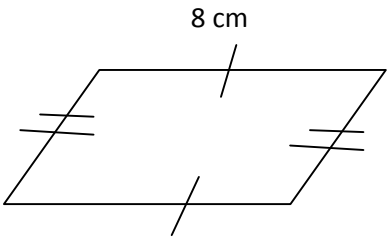
b)



15 + 5 = 20 cm

30 – 20 = 10 cm

c)

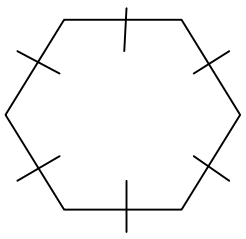


Two equal sides are 8 cm each = 16 cm

The other two equal sides are 30 – 16 = 14 cm

Each side is 7 cm

d) A



6 equal sides are 30cm; each side is 5 cm

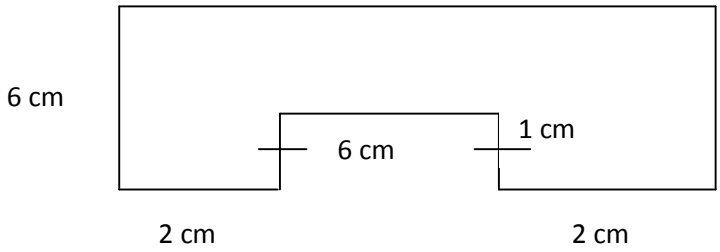
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3) A soccer field is 100 metres long and 30 metres wide. How far would you walk if you went twice around it?

100 + 30 + 100 + 30 = 260 m perimeter

260 x 2 = 520 m for twice around the pitch

4) Calculate the perimeter of the following shape



The length of the longest side = 2 + 6 + 2 = 10

The perimeter is 6 + 10 + 6 + 2 + 1 + 6 + 1 + 2 = 34 cm

5) Two ants walk around a square. They start at the same corner at the same time. The first ant goes round the square twice while the second ant goes around once. In total they travelled 36 metres, what is the length of each side of the square?

3 times around the square = 36 metres

Perimeter of square = 12 metres

Length of one side = 3 metres

6) What effect does doubling the length and width of a square have on its perimeter?

Doubles the perimeter

7) What effect does doubling the length of a rectangle while keeping the width the same have on its perimeter?

Adds two times the original length to the perimeter

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8) What must the side length of an equilateral triangle be so it has the same perimeter as a square of side length 12 cm?

Perimeter of square = 4 x 12 = 48 cm

Length of each side of triangle = 48 ÷ 3 = 16 cm

9) The perimeter of a rectangle is 40 cm. If it is 6 cm wide, what is its length?

Perimeter = length + width + length + width

40 = length + length + 6 + 6

Length + length = 28

Length = 14 cm

10) The length of a rectangle is 4 cm more than its width. If the perimeter of the rectangle is 16 cm, what are its measurements?

Perimeter = 16 = length + width + length + width

16 = (width + 4) + width + (width + 4) + width

16 = width + width + width + width + 8

Width + width + width + width = 8

Width = 2 cm

11) Five pieces of string are placed together so they form a regular pentagon. Each piece of string is 8 cm long. How long should the pieces of string be to make a square having the same perimeter as the pentagon?

Perimeter of pentagon = 5 x 8 = 40 cm

Each side of square should be 40 ÷ 4 = 10 cm

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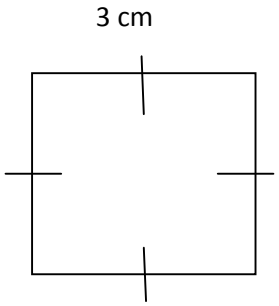
Exercise 3

Simple Area Problems

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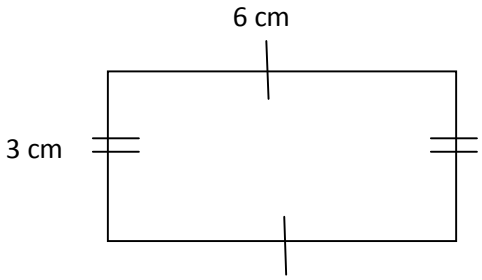
1) Calculate the area of the following

a)



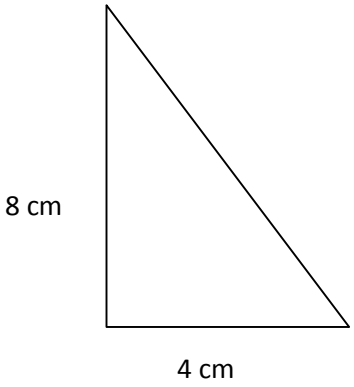
Area of square = length x width = 3 x 3 = 9 cm^2

b)



Area of rectangle = length x width = 6 x 3 = 18 cm^2

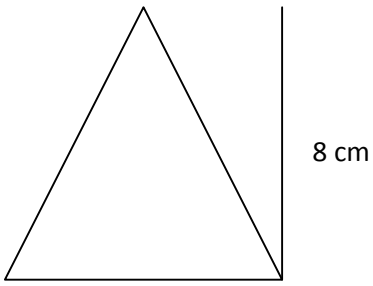
c)



Area of triangle = (base x height) x ½ = 4 x 8 x ½ = 16 cm^2

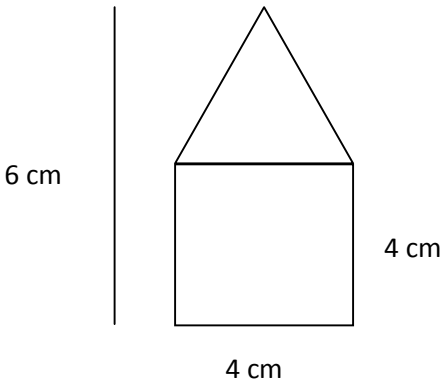
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d)



Area of triangle = (base x height) x ½ = 8 x 8 x ½ = 32 cm^2

e)



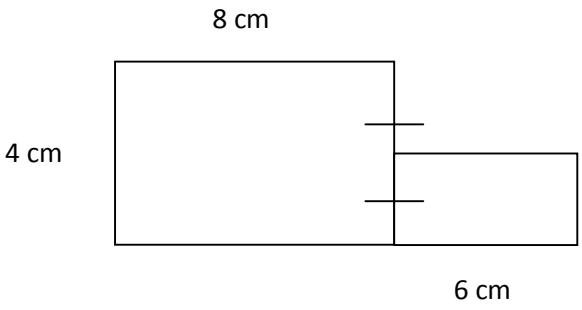
Height of triangle = 6 – 4 = 2 cm

Area of triangle = 4 x 2 x ½ = 4 cm^2

Area of square = 4 x 4 = 16 cm^2

Total area = 4 + 16 = 20 cm^2

f)



Area of large rectangle = 4 x 8 = 32 cm^2

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Area of small rectangle = 6 x 2 = 12 cm^2

Total area = 32 + 12 = 44 cm^2

2) A park measures 200 metres long by 50 metres wide. What is the area of the park?

Area = 200 x 50 = 10,000 m^2

3) The floor of a warehouse is 18 metres long and 10 metres wide. One can of floor paint covers 45 square metres. How many cans of paint are needed to paint the floor?

Area = 18 x 10 = 180 m^2

180 ÷ 45 = 4 cans of paint

4) A tablecloth is 2 metres long and 500 cm wide. What is its area?

500 cm = 0.5 m

Area = 2 x 0.5 = 1 m^2

5) A wall measures 2.5 metres high by 6 metres wide. A window in the wall measures 1.5 metres by 3 metres. What area of the wall is left to paint?

Area of wall = 2.5 x 6 = 15 m^2

Area of window = 1.5 x 3 = 4.5 m^2

Area to paint = 15 – 4.5 = 10.5 m^2

6) A customer requires 60 square metres of curtain fabric. If the width of a roll is 1.5 metres, what length of fabric does he require?

Area = length x width

60 = length x 1.5

So length needed = 40 m^2

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7) A square piece of wood has an area of 400 square centimetres. How long and how wide is it?

Need a number that is multiplied by itself to get 400

That is we need the square root of 400 = 20 m

8) A stretch of road is 5 km long and 4 metres wide. What is its area?

5 km = 5000 m

Area = 5000 x 4 = 20,000 m^2

9) A table is 400 centimetres long and 80 centimetres wide. What is its area in square metres?

400 cm = 4 m

80 cm = 0.8 m

Area = 4 x 0.8 = 3.2 m^2

10) A car park is 2.5 km long and 800 metres wide. What is its area in square metres and square kilometres?

2.5 km = 2500 m

Area = 2500 x 800 = 2,000,000 m^2

2,000,000 m^2 = 2 km^2

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11) Investigate the areas of rectangles that can be made using a piece of string that is 16 cm long. Complete the following table to help you. (Use whole numbers only for lengths of sides)

Length (cm)	Width (cm)	Area (cm^2)
1	7	7
2	6	12
3	5	15
4	4	16
5	3	15
6	2	12
7	1	7

12) A farmer has 400 metres of fencing in which to hold a horse. He wants to give the horse as much grazing area as possible, while using up all the fencing. Using your answers to question 11 as a guide, what should the length and width of his enclosure be, and what grazing area will the horse have?

From the answer to question 11 (and question 7), the greatest area is gained when the sides are equal (a square). Therefore if the farmer has 4 sides each 100 metres he will have the greatest possible area.

The area will be 100 x 100 = 10,000 m^2

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Year 5 Mathematics

Measurement:
Volume & Capacity

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Exercise 1

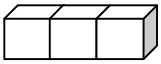
Determining Volume From Diagrams

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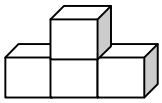
1) Each cube in the following diagrams has a volume of $1cm^3$. Calculate the volume of the structure.

a) 

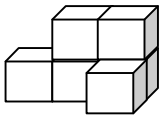
1 cm^3

b) 

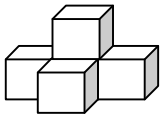
3 cm^3

c) 

4 cm^3

d) 

6 cm^3

e) 

5 cm^3

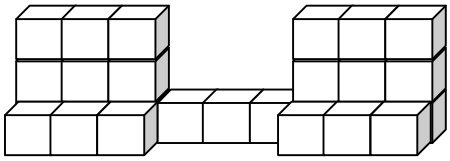
2) A wall is 5 blocks long, 3 blocks wide and 2 blocks high. Each block has a volume of $1m^3$. How many blocks are in the wall? What is the volume of the wall?

A diagram will assist you

A correct diagram should show 30 blocks, volume of wall is 30 m^3

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3) Each block in the following diagram has a volume of 0.5 cm^3, what is the volume of the structure?



There are 12 blocks on each side and 3 in the middle, for a total of 27 blocks

Volume = 27 x 0.5 = 13.5 cm^3

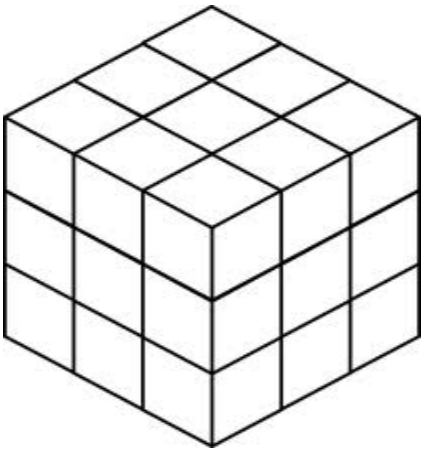
4) The image below shows a chessboard; each square is a piece of wood that has a volume of 50 cm^3. Ignoring the border, what is the volume of the chessboard?



There are 64 small squares on a chessboard

Volume = 64 x 50 = 3200 cm^3

5) Each small cube that makes up the large one has a volume of 1 cm^3. What is the total volume of the large cube?



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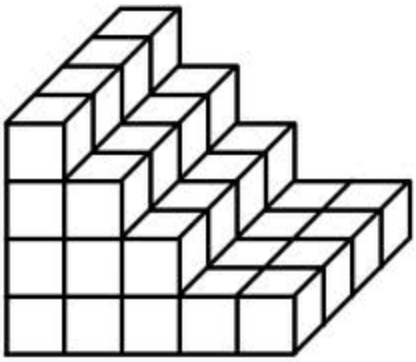
There are 27 small cubes; therefore the volume is 27 cm^3

Use your result to show the general method of calculating the volume of a large cube.

27 = 3 x 3 x 3, which is the length x width x height

This method can be applied to any large cube if the sizes of the sides can be calculated from sizes of smaller blocks, or is given similarly to area

6) Each cube in the image below has a volume of 1 cm^3. What is the volume of the structure?



There are 16 blocks in the back row, 12 in the next, 8 in the next, and 4 in the last, for a total of 40 blocks

The volume of the structure is 40 cm^3

7) What is the volume of a stack of bricks each having a volume of 900 cm^3 if they are stacked 4 high, 5 deep, and 7 wide?

There are 4 x 5 x 7 = 140 bricks in the stack

Volume = 900 x 140 = 126,000 cm^3

8) Three hundred identical cubes are made into a wall that is 3 blocks high, 5 blocks wide and 20 blocks long. If the total volume of the wall is 8,100,000 cm^3, what is the length of each side of one cube?

There are 3 x 5 x 20 = 300 cubes in the wall

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If the total volume of the wall is 8,100,000 then the volume of each cube is:

$8{,}100{,}000 \div 300 = 27000\ cm^3$

$30 \times 30 \times 30 = 27000$

Since the sides of the small cubes must all be the same length, each side must be 30 cm long

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Exercise 2

Units of Measurement: Converting & Applying

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1) Convert the following to cm^3

a) 1000 mm^3

1 cm^3

b) 1 m^3

1,000,000 cm^3

c) 2000 mm^3

2 cm^3

d) 3500 mm^3

3.5 cm^3

e) 0.1 m^3

100,000 cm^3

2) Convert the following to m^3

a) 1,000,000 cm^3

1 m^3

b) 2,000,000 cm^3

2 m^3

c) 1 km^3

1,000,000,000 m^3

d) 0.1 km^3

100,000,000 m^3

e) 100,000 cm^3

0.1 m^3

3) A box has the measurements 100 mm x 100 mm x 10 mm. What is the volume of the box in cm^3?

Volume = 100 x 100 x 10 = 100,000 mm^3 = 100 cm^3

4) A sand pit measures 400 cm x 400 cm x 20 cm. How many cubic metres of sand should be ordered to fill it?

Volume = 400 x 400 x 20 = 3,200,000 cm^3 = 3.2 m^3

5) Chickens are transported in crates that are stacked on top of and next to each other, and then loaded into a truck. Each crate has a volume of approximately 30000 cm^3.

How many crates could fit inside a truck of volume:

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a) 300000 cm^3

300,000 ÷ 30,000 = 10 crates

b) 30 m^3

30 m^3 = 30,000,000 cm^3

30,000,000 ÷ 30,000 = 1,000 crates

c) 270 m^3

270 m^3 = 9 x 30 m^3, so using the answer to part b, the truck would hold 9 x 1000 = 9000 crates

6) A hectare is equal to 10,000 m^2. How many hectares in 1 km^2?

1 km^2 = 1,000,000 m^2

1,000,000 ÷ 10,000 = 100 hectares

7) Put the following volumes in order from smallest to largest

10 m^3, 0.1 km^3, 5,000,000 cm^3, 10,000 mm^3

Change all to m^3

10 m^3, 100,000 m^3, 5 m^3, 0.00001 m^3

Order is 4, 3, 1, 2

8) Put the following in order from largest to smallest

100 cm^3, 10,000 mm^3, 0.01 m^3, 10 cm^3

Change all to cm^3

100 cm^3, 10 cm^3, 10,000 cm^3, 10 cm^3

Order is 3, 1, 2 and 4 equal

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9) A cube has a side length of 2000 mm. What is its volume in cm^3 and in m^3?

2000 mm = 200 cm = 2 m

Volume = 200 x 200 x 200 = 8,000,000 cm^3 = 8 m^3

Volume = 2 x 2 x 2 = 8 m^3

Conversion to appropriate units can be done before or after calculation of volume

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Exercise 3

Relationship Between Volume & Capacity

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1) Convert the following to cm^3

a) 1 mL

1 cm^3

b) 100 mL

100 cm^3

c) 350 mL

350 cm^3

d) 2 L

2 L = 2000 mL = 2000 cm^3

e) 10 L

10 L = 10,000 mL
= 10,000 cm^3

f) 4.2 L

4.2 L = 4200 mL = 4200 cm^3

2) Convert the following to Litres

a) 1500 cm^3

1500 cm^3 = 1500 mL = 1.5 L

b) 500 cm^3

500 cm^3 = 500 mL = 0.5 L

c) 1250 cm^3

1250 cm^3 = 1250 mL
= 1.25 L

d) 10,000 cm^3

10,000 cm^3 = 10,000 mL
= 10 L

e) 100 cm^3

100 cm^3 = 100 mL = 0.1 L

3) The following questions show the side length of a cube. Calculate the capacity of each cube in Litres

a) 10 cm

V = 10 x 10 x 10 = 1000 cm^3

1000 cm^3 = 1000 mL = 1 L

b) 100 cm

V = 100 x 100 x 100
= 1,000,000 cm^3

1,000,000 cm^3 = 1,000,000 mL = 1,000 L

c) 500 cm

500 x 500 x 500 =
125,000,000 cm^3

125,000,000 cm^3 =
125,000,000 mL

= 125,000 L

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d) 1000 cm

$V = 1000 \times 1000 \times 1000$
$= 1{,}000{,}000{,}000\ cm^3$

$1{,}000{,}000{,}000\ cm^3$
$= 1{,}000{,}000{,}000\ mL$

$= 1{,}000{,}000\ L$

4) The following questions show the capacity of a cube in Litres. What is the side length of the cube?

a) 1

1 L = 1000 mL

$1000\ mL = 1000\ cm^3$

Each side is 10 cm

b) 8

8 L = 8,000 mL

$8{,}000\ mL = 8{,}000\ cm^3$

Each side is 20 cm

c) 27

27 L = 27,000 mL

$27{,}000\ mL = 27{,}000\ cm^3$

Each side is 30 cm

d) 1000

1000 L = 1,000,000 mL

$1{,}000{,}000\ mL = 1{,}000{,}000\ cm^3$

Each side is 100 cm

5) Convert the following to Litres

a) $5\ m^3$

5,000 L

b) $10\ m^3$

10,000 L

c) $7.5\ m^3$

7,500 L

d) $3.52\ m^3$

3520 L

e) $0.1\ m^3$

100 L

6) Convert the following to m^3

a) 500 L

$0.5\ m^3$

b) 800 L

$0.8\ m^3$

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c) 3000 L

$3\ m^3$

d) 10,000 L

$10\ m^3$

e) 1550 L

$1.55\ m^3$

7) A swimming pool is 50 metres long by 10 metres wide, and has an average depth of 2 metres. What is the capacity of the pool in litres?

Volume = 50 x 10 x 2 = $1000\ m^3$

$1000\ m^3$ = 1,000,000 L

8) A swimming pool has a capacity of 500,000 litres. If it is 100 metres long by 5 metres wide, what is its average depth?

500,000 L = $500\ m^3$

100 x 5 x depth = 500

Therefore average depth = 1 m

9) A water tank is 10 metres long by 8 metres wide by 10 metres deep. A chemical has to be added at the rate of one tablet per 200,000 litres. How many tablets need to be added to the tank?

Volume = 10 x 8 x 10 = $800\ m^3$

Capacity = 800,000 L

Therefore 4 tablets are needed

10) Petrol sells for $1.50 per litre. A tanker carried $300,000 worth of petrol. The tanker was in the shape of a rectangular prism and measured 5 metres long and 4 metres deep. How long was the tanker?

300,000 ÷ 1.50 = 200,000 Litres of petrol

200,000 L = $200\ m^3$

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5 x 4 x length = 200 m^3

Therefore the tanker was 10 m long

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Year 5 Mathematics

Mass & Time

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Exercise 1

Units of Mass Measurement:

Converting & Applying

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1) Convert the following to kilograms

a) 1000 g

1 kg

b) 2000 g

2 kg

c) 2500 g

2.5 kg

d) 500 g

0.5 kg

e) 750 g

0.75 kg

f) 1.5 Tonne

1500 kg

g) 4 Tonne

4 000 kg

2) Convert the following to grams

a) 1000 mg

1 g

b) 3000 mg

3 g

c) 2 kg

2000 g

d) 3.5 kg

3,500 g

e) 600 mg

0.6 g

f) 100 mg

0.1 g

g) 100 kg

100,000 g

3) Convert the following to milligrams

a) 4 g

4000 mg

b) 10 g

10,000 mg

c) 0.2 g

200 mg

d) 1 kg

1,000,000 mg

e) 100 g

100,000 mg

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4) A man places four 750 gram weights on one side of a scale. How many 1 kg weights must he place on the other side of the scale for it to balance?

4 x 750 g = 3 kg

He must place 3 x 1 kg weights on the other side

5) Meat is advertised for $20 per kilogram. How much would 250 grams of the meat cost?

250 g = 0.25 kg

0.25 x $20 = $5

6) A rock collector collects 5 rocks. They weigh 300 grams, 400 grams, 500 grams, 1.5 kilograms, and 2 kilograms respectively. What was the total weight of his collection in grams and in kilograms?

300 + 400 + 500 + 1500 + 2000 g = 4700 g = 4.7 kg

7) A vitamin comes in tablets each of which has a mass of 200 milligrams. If there are 500 tablets in a bottle, and the bottle has a mass of 200 grams, what is the total weight of the bottle of tablets in grams and in kilograms?

200 x 500 = 100,000 mg of tablets

100,000 mg = 100 g of tablets

Bottle plus tablets = 200 g + 100 g = 300 g = 0.3 kg

8) John has a parcel of mass 1.5 kilograms to send by courier. Courier company A charges $15 per kilogram, while courier company B charges 1.5 cents per gram. Which courier company is cheaper and by how much?

1.5 kg x 15 = $22.50

1.5 kg = 1500 g

1500 g x .15 = $225

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Company A is cheaper

9) Which has more mass and by how much? Two hundred balls each with a mass of 100 grams, or 50 balls each with a mass of 0.5 kilograms.

200 x 100 g = 20,000 g = 20 kg

50 x 0.5 kg = 25 kg

The second amount is heavier by 5 kg

10) A mixture has the following chemicals in it

- 1 kg of chemical A
- 750 g of chemical B
- 300 g of chemical C
- 800 mg of chemical D
- 700 mg of chemical E
- 500 mg of chemical F

What is the total mass of the mixture in kilograms, grams, and milligrams?

Change all to grams: 1000 + 750 + 300 + 0.8 + 0.7 + 0.5 = 2052 g = 2.052 kg

= 2,052,000 mg

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Exercise 2

Estimating Mass

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1) For each of the following, state whether the usual unit of mass measurement is mg, g, kg, or tonnes

a) A human

kg

b) Packet of lollies

g

c) An elephant

Tonnes

d) Loaf of bread

g

e) Paper clip

g

f) A car

Tonnes

g) An ant

mg

2) A jack has a lifting capacity of 200 kg. Which of the following could be safely lifted by the jack?

- A truck
- A pool table
- A barbeque
- A spare tyre
- A carton of soft drink

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Carton of drinks, spare tyre, barbeque

3) Alfred buys a carton of butter that contains 10 x 375 gram tubs. What is the approximate mass of the carton to the nearest kilogram?

10 x 375 g = 3750 g = 3.75 kg ≈ 4 kg

4) If a person rode on or in each of the following, for which would they increase the mass greatly?

- Horse
- Skateboard
- Bicycle
- Car
- Airplane
- Roller skates

Skateboard, bicycle, roller skates

5) A car and a truck travelling the same speed each hit the same size barrier. Which one would push the barrier the furthest?

The truck since it has more mass

6) Put the following balls in order from smallest to heaviest mass

- Medicine ball
- Table tennis ball
- Tennis ball
- Golf ball
- Football
- Bowling ball

Table tennis ball, golf ball, tennis ball, football, bowling ball, medicine ball

Note: a golf ball and tennis ball are similar in weight

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7) Approximately how many average mass adults could fit into a boat with a load limit of 1 tonne

An average adult weighs approximately between 75 and 85 kg

1000 ÷ 75 = 13.33

1000 ÷ 85 = 11.76

12 to 13 adults could fit in the lifeboat

Note to tutor: the exact answer is not important, but correct estimation and hence ball park figure is necessary

8) Which has more mass; a kilogram of feathers or a kilogram of bricks? Explain your answer

Since they both weigh 1 kg, they weigh the same.

What is true is that there are far more feathers in a kg than there are bricks

We say that feathers are less dense than bricks

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Exercise 3

Notations of Time

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1) Which of the following activities usually occur AM and which usually occur PM?

- Waking from a night's sleep

  AM

- Having dinner

  PM

- Going to school

  AM

- Having lunch

  PM

- Sport training

  PM

- Watching the sunset

  PM

- People working

  AM and PM

2) School starts for Joseph at 9 AM and goes for 4 hours until lunchtime. At what time (AM or PM) does Joseph eat his lunch?

1 PM

3) Write the time including AM or PM at one minute past midnight

12:01 AM

4) Convert the following to AM or PM notation

a) 1030

10:30 AM

b) 1115

11:15 AM

c) 1515

3:15 PM

d) 0200

2:00 AM

e) 1600

4:00 PM

f) 2120

9:20 PM

g) 0725

7:25 AM

h) 1925

7:25 PM

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5) Convert the following to 24 hour time notation

a) 3:00 PM

1500

b) 1:15 AM

0115

c) Midnight

0000

d) 10:45 PM

2245

e) 7:55 PM

1955

f) Noon

1200

6) Put the following times in order from earliest to latest

- 1515
- 3:10 AM
- 4:20 PM
- 1600
- 2020
- 11:22 AM

3:10 AM, 11:22 AM, 1515, 1600, 4:20 PM, 2020

7) Charlie went to bed at 8:30 PM, Andrew went to bed at 1950, and Peter went to bed at 2040. Who went to bed earliest and who went to bed latest?

Andrew (1950 = 7:50 PM)

Peter (2040 = 8:40 PM)

8) In Antarctica on the 7th December 2011, the sun rose at 0106 and set at 2351. Convert these times to AM and PM notation. What does your answer reveal to you?

0106 = 1:06 AM

2351 = 11:51 PM

It was daylight for virtually the whole 24 hours

9) Three people wrote down the following statements

- "I eat dinner at about 6 o'clock every evening"
- "I eat dinner at about 0715 every evening"
- "I eat dinner at about 1925 every evening"

Who was likely to have used the wrong time notation?

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The second person, since 0715 is 7:15 AM

10) Three people are catching plane flights from the same airport on the same day. Andrew's flight leaves at 2:30 in the morning. Bob's flight leaves at 1510, and Chris' flight leaves at 2:58 PM. If check in is three hours before takeoff, who would have to arrive at the airport when their watch read AM time?

Andrew's watch would read 11:30 PM

Bob's would read 12:10 PM

Chris' would read 11: 58 AM

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Exercise 4

Elapsed Time, Time Zones

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1) How much time is there between the following pairs of times?

a) 1:15 AM and 7:20 AM

6 hours and 5 minutes

b) 4:35 PM and 8:50 PM

4 hours and 15 minutes

c) 9:12 PM and 11:59 PM

2 hours and 47 minutes

d) 4:25 AM and 6:40 PM

14 hours and 15 minutes

e) 11:44 AM and 6:51 PM

7 hours and 7 minutes

f) Noon and 3: 22 PM

3 hours and 22 minutes

2) How much time is there between the following pairs of times?

a) 0312 and 1133

8 hours and 21 minutes

b) 1533 and 1748

2 hours and 15 minutes

c) 1614 and 2217

6 hours and 3 minutes

d) 0830 and 1435

6 hours and 5 minutes

e) 1040 and 1853

8 hours and 13 minutes

f) 0958 and 1459

5 hours and 1 minute

3) How much time is there between the following pairs of times?

a) 6:45 AM and 10:16 AM

3 hours and 31 minutes

b) 9:30 PM and 11:11 PM

1 hour and 41 minutes

c) 2:18 AM and 4:17 AM

1 hour and 59 minutes

d) 5:23 AM and 2:18 PM

8 hours and 55 minutes

e) 7:26 PM and 3:07 AM

7 hours and 41 minutes

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f) 11:05 PM and 9:02 AM

9 hours and 57 minutes

4) How much time is there between the following pairs of times?

a) 0415 and 2:20 PM

10 hours and 5 minutes

b) 6:35 AM and 1543

9 hours and 8 minutes

c) 2120 and 2:25 AM

5 hours and 5 minutes

d) 0333 and 3:23 PM

11 hours and 50 minutes

e) 11:12 AM and 1601

4 hours and 49 minutes

f) 1117 and 3:07 AM the next day

15 hours and 50 minutes

5) A bus timetable states that bus number 235 leaves at 1525 and that the service runs every 35 minutes after that. What are the times of the next three buses (in 24 hour notation)?

1600, 1635, 1710

6) Andre has to catch a train and a bus to get home. His train leaves at 1610, and arrives at the bus station at 5:05 PM. He waits ten minutes and catches the bus which takes 43 minutes to reach his stop. He then walks home for 5 minutes. How long does his journey take, and what time does he arrive home (Answer in both Pm and 24 hour notation)

1610 (4:10 Pm) to 5:05 PM = 55 minutes

10 minute wait

43 minutes by bus

Walk for 5 minutes

Total time for journey to home = 1 hour 53 minutes

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Arrives home at 1803; 6:03 PM

7) The table below shows the time difference between some cities of the world.

City	Time difference (from Sydney)	Local time
Auckland	+ 2 hours	0900
Sydney	0 hours	0700
Hong Kong	-3 hours	0400
Paris	-8 hours	2300
London	-11 hours	2000
New York	-16 hours	1500
Los Angeles	-19 hours	1200

8) Perth summer time is three hours behind Sydney summer time. A plane leaves Sydney at 1400 Sydney time. The flight takes 4 and one half hours. What is the time in Perth when the flight lands?

When the plane leaves Sydney it is 1400 – 3 hours = 1100 in Perth

1100 + 4.5 hours = 1530

9) From the table in question 7, if it is 4 PM on New Year's Eve in Los Angeles, what is the time and day in Sydney?

Sydney = LA + 19 hours

4 PM + 19 hours = 11 AM New Year's Day

10) A man boards a flight in New York at 10 PM. The flight takes 7 hours to reach London. Using the table in question 7 as a guide, what time is it in London when the plane lands?

Difference between London and New York = 5 hours

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10 PM New York = 3 AM London

3 AM + 7 hours = 10 AM in London when plane lands

11) The circumference of the Earth at the equator is approximately 40070 km. Auckland and Paris are 12 hours apart in time. Using the knowledge that the Earth takes approximately one day (24 hours) to rotate once on its axis:

a) What is the approximate distance from Auckland to Paris?

12 hours equals half a day approximately

In 12 hours the equator has turned through ½ of 40070 km

Therefore the cities are approximately 20,000 km apart

b) (Challenge Question): What is the approximate speed of the rotation of the earth in Kilometres per hour?

40070 km in 24 hours = 40070 ÷ 24 = 1670 km per hour approximately

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Year 5 Mathematics

Space

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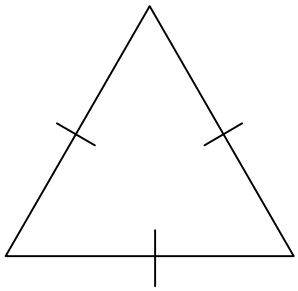
Exercise 1

Types & Properties of Triangles

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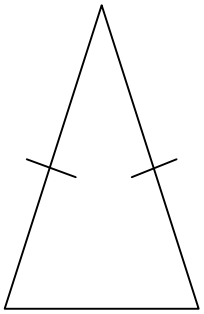
1) Name the following triangles

a)



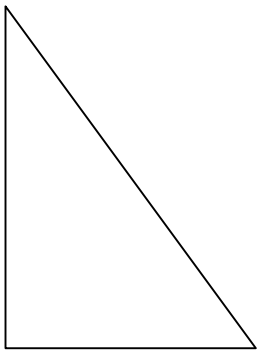
Equilateral triangle

b)



Isosceles triangle

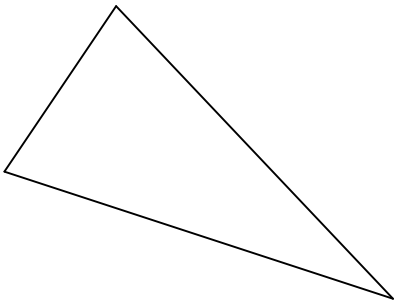
c)



Right-angled triangle

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d)



Scalene triangle

2) True or false? The three angles of an isosceles triangle are congruent (the same size)

False; only two of the angles are congruent

3) Which **types** of triangle can have two of its three sides equal?

Isosceles & Right-angled

4) Which type of triangle has two angles that are equal to 90 degrees?

No triangle can have two angles of 90 degrees

5) Name two unique characteristics of an equilateral triangle

All angles are congruent

All sides are congruent

6) How many sides of an isosceles triangle are equal in length?

two

7) A triangle that has no sides equal in length is either a _____ triangle or a _____- triangle

Scalene, Right-angled

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8) If a square is cut across from one diagonal to another what type(s) of triangle(s) are formed?

Isosceles

9) If a rectangle is cut across from one diagonal to another what type(s) of triangle(s) are formed?

Scalene

10) What is the size of each angle of an equilateral triangle?

60 degrees

11) If one of the angles of a right-angled triangle measures 60 degrees, what are the sizes of the other two angles?

90 degrees and 30 degrees

12) Which type(s) of triangle(s) can have an angle greater than 90 degrees

Equilateral cannot since its angles are all 60 degrees

Right-angled cannot since one angle must be 90 degrees and the other two must be less

Isosceles can as long as the other two angles are equal (e.g. 140, 20, 20)

Scalene can

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Exercise 2

Types & Properties of Quadrilaterals

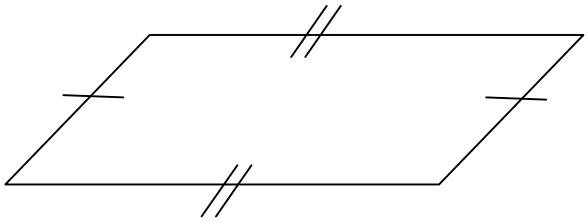
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1) How many sides does a quadrilateral have?

4

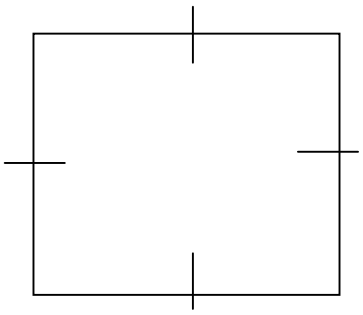
2) Name the following types of quadrilaterals

a)



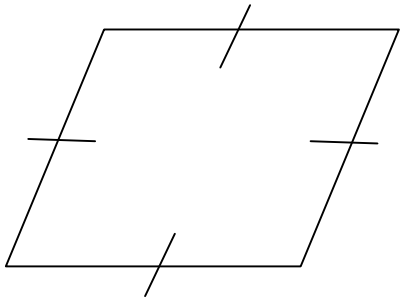
Parallelogram

b)



Square

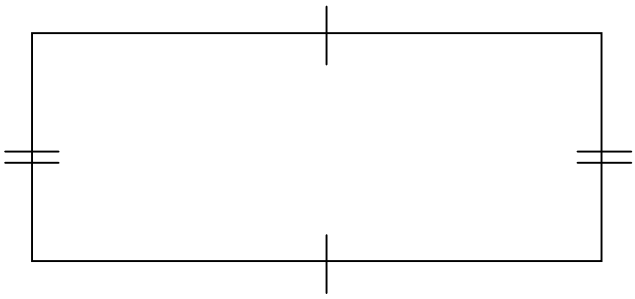
c)



Rhombus

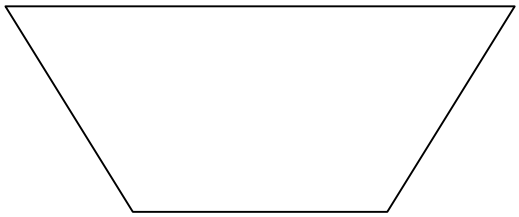
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d)



Rectangle

e)



Trapezoid

3) Each angle of a square is _____ degrees

90

4) Name three quadrilaterals that have angles of more than 90 degrees

Parallelogram
Rhombus
Trapezoid

5) Name a quadrilateral that has a pair of sides not parallel

Trapezoid

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6) A rhombus is a special type of _____

Parallelogram

7) A square is a special type of _____

Rectangle

8) Name three characteristics that are shared by a square and a rectangle

Two sets of parallel sides

All angles are 90 degrees

4 sided

9) Name two characteristics that are shared by a trapezoid and a rectangle

4 sided

One pair of parallel sides

10) Name the quadrilateral(s) that can have angles greater than 90 degrees

Rhombus
Parallelogram
Trapezoid
Kite (not looked at here)

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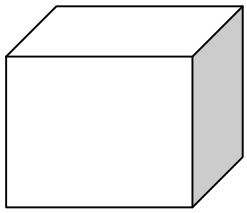
Exercise 3

Prisms & Pyramids

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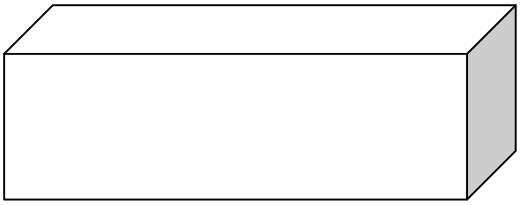
1) Name each of the following shapes

a)



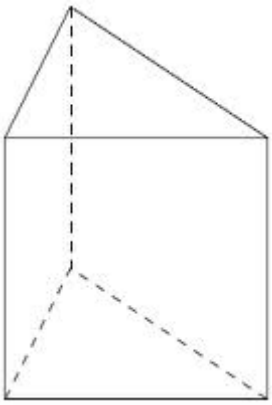
Cube

b)



Rectangular prism

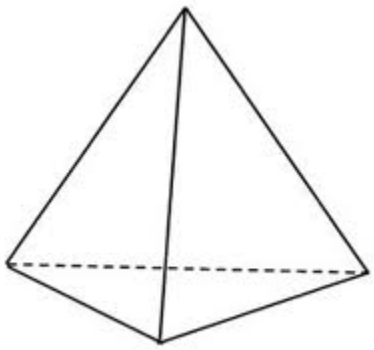
c)



Triangular prism

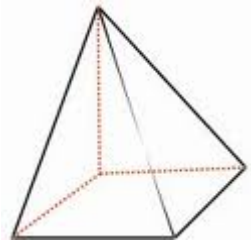
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d)



Triangular pyramid

e)



Square pyramid

2) What is the major difference between prisms and pyramids?

Pyramids have triangular sides that join at an apex, prisms have rectangular sides and join the base and top which are the same shape

3) A shape has a hexagon at each end and rectangular sides joining them. What is this shape called

Hexagonal prism

4)

a) How many faces does a rectangular prism have?

6

b) How many edges does a rectangular prism have?

12

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c) How many vertices (corners) does a rectangular prism have?

8

5)

a) How many faces does a triangular pyramid have?

4

b) How many edges does a triangular pyramid have?

6

c) How many vertices does a triangular pyramid have?

4

6)

a) How many faces does a triangular prism have?

5

b) How many edges does a triangular prism have?

9

c) How many vertices does a triangular prism have?

6

7) From your answers to questions 4 to 6, is there a rule that connects the number of faces, edges and vertices in a prism or pyramid?

Yes; number of faces + number of vertices = number of edges +2

8) All prisms have at least _____ pair of parallel faces

One; the base pairs are always parallel

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9) Pyramids have _____ pairs of parallel faces

Zero

10) What is the main feature of a cube that distinguishes it from other prisms?

All sides are the same length

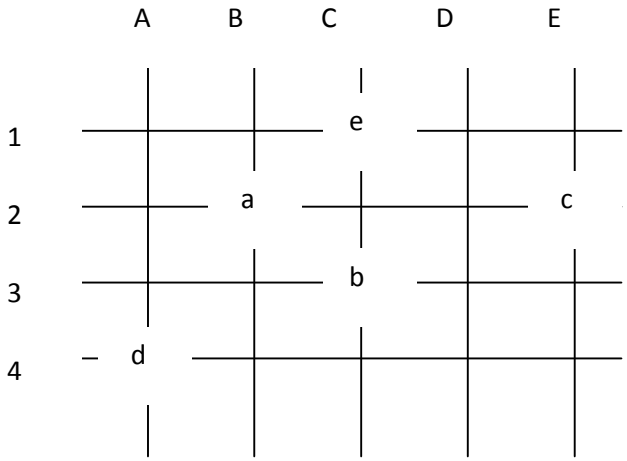
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Exercise 4

Maps: Co-ordinates, Scale & Routes

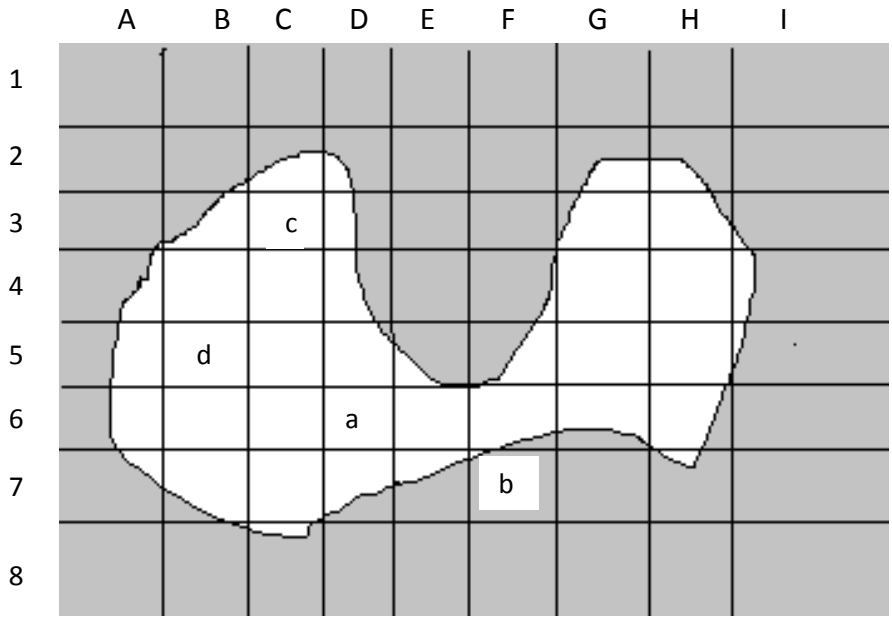
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1) Using the grid below, write the co-ordinates of the points a to e



a = B2, b = C3, c = E2, d = A4, e = C1

2)



Mark the following co-ordinates on the map

a) D6

b) F7

c) C3

d) B5

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e) If the white portion of the map represents land and the grey represents water, give the co-ordinates of a square:

I. That is all land

G5 is an example

II. That is all water

C1 is an example

III. That is approximately half land and half water

B3 is an example

IV. That is mostly land

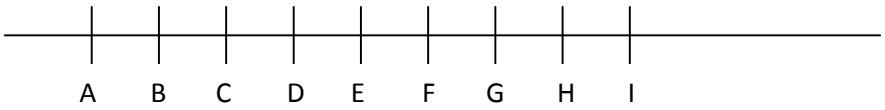
D5 is an example

V. That is mostly water

E5 is an example

NOTE there are many more

3)



The distance between each mark on the line represents 50 km. What distance is represented from:

a) A to D

150 km

b) B to E

150 km

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c) B to G

250 km

d) H to C

250 km

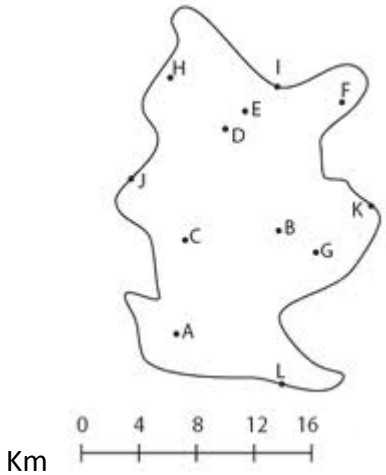
e) A to F and back to D

350 km

f) G to C and back to E

300 km

4) Use the map and scale below it to answer the questions



What are the distances from:

a) Points A and H

16 km

b) Points C and K

12 km

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c) Points F and D

8 km

d) Points B and G

2km

e) Points L and K

12 km

5) The map below shows the Murray River and the south eastern portion of Australia



a) What is the approximate distance from Brisbane to Sydney?

600 km

b) What is the approximate distance from Canberra to Melbourne?

450 km

c) Approximately how long is the border between New South Wales and Queensland?

1100 km

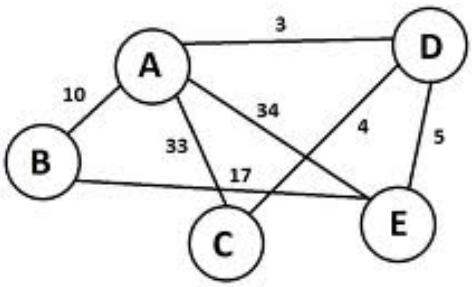
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d) By treating the state of New South Wales as a rectangle, estimate its area.

Length 1100 km (the border), average width 750 km

Area approximately 1100 x 750 = 825,000 km^2

6)



The diagram shows the shortest distance between any two points

a) Along which path or paths is the shortest distance from A to E?

A to D to E = 8 km

b) What is the shortest distance from B to C?

B to A to D to C = 17 km

c) What is the shortest distance from D to E if you must also go through point A?

D to A to B to E = 30 km

d) What is the shortest distance if you must start at point A, visit each point once but only once and return to point A?

Either ABEDCA or ACDEBA; both 69 km

7) Draw a scale map that has the following information

a) A scale of 1 cm equals 10 km

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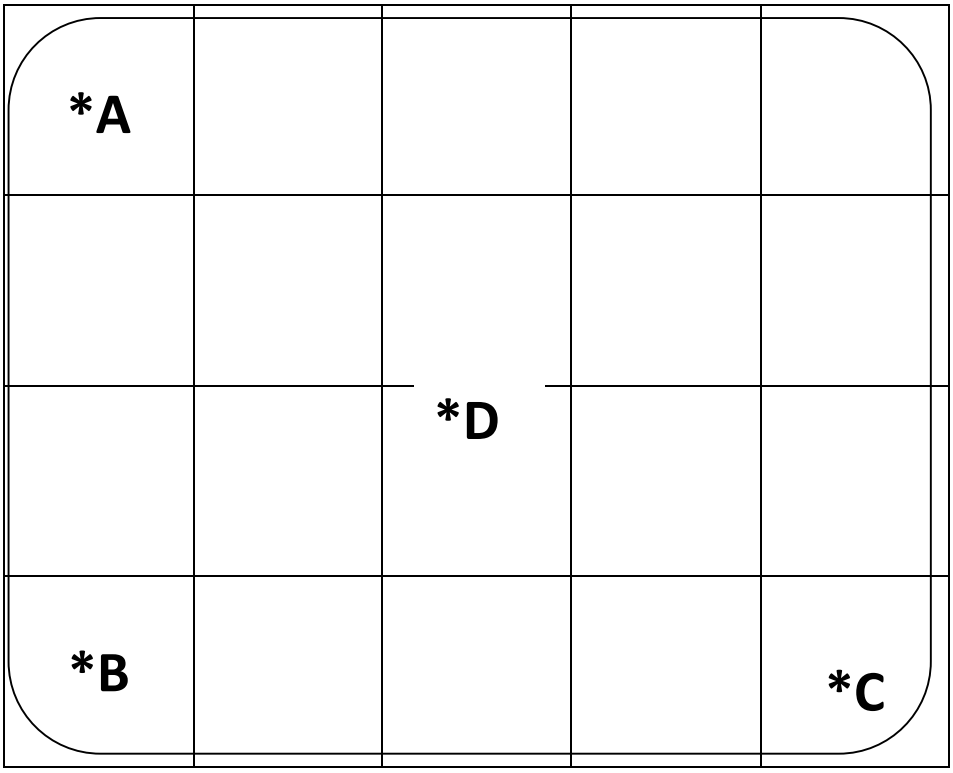
b) The distance from A to B is 30 km

c) Point B is located at co-ordinate A5

d) The distance from point A to point C is 50 km, but is 70 km if you go via point B

e) Point D is an equal distance (25 km) from points A and C

f) The points all lie on an island that is in the approximate shape of a rectangle and has an area of 2000 km^2



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